Simple local trace formulas for unramified p-adic groups

Revised, corrected and updated

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Abstract

Let G be a connected unramified reductive group over a p-adic field F, K a hyperspecial maximal compact subgroup, f a spherical function, $f^\vee$ its Satake transform and ϕ a smooth compactly supported function on the regular elliptic set. Following the method of Waldspurger [Wa89a] for $GL(N)$ (truncation, Casselman's asymptotics, Macdonald's formula), we study the spectral expansion

$$\int_{G(F)^1\times G(F)^1} f(h)\phi(g^{-1}hg)\,dg\,dh = \int f^\vee(\chi)\,dI_\phi(\chi),$$

equivalently of elliptic orbital integrals $\int f(g^{-1}\gamma g)\,dg$ of spherical functions, as an integral over the compact torus of unitary unramified characters. For a truncation parameter T the identity holds exactly with an explicit truncated kernel $I^T(\chi,\phi)$, and I^T is a finite sum of vertex terms $\psi(T_Q)p(T_Q)$ (Brion's decomposition), whose T-independent part $I(\chi,\phi)$ is a rational function of χ. Under a regularity hypothesis on the vertex terms the identity holds with $dI_\phi = I(\chi,\phi)\,d\mu(\chi)$, $d\mu$ the Plancherel measure; the hypothesis is verified for $PGL(2)$. For $SL(2)$ it fails, and the spectral measure has an atom at the unramified quadratic character, the non-regular unitary character where the Plancherel density does not vanish; the atom is the κ-orbital integral, as the fundamental lemma predicts. This corrects the claim of the 1997 version that $I(\chi,\gamma)$ is always a rational function. We also give the κ-twisted version for $SL(n)$ in the light of the now-proved fundamental lemma, worked examples for $PGL(2)$ and $SL(2)$ checked numerically on the Bruhat–Tits tree, a sketch of the derivation from Arthur's local trace formula, a literature review and a glossary.

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*Claude is an AI model made by Anthropic. This revision starts from a corrected version of the paper circulated in October 1997. Changes from the 1997 text are listed in the file `CHANGES.md` that accompanies this paper; substantive changes, and everything not fully verified, are flagged in the text by a dated "Remark 2026.n". The literal transcription of the 1997 version, its errata log and a referee report are in `../original-1997/`.

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About this revision

History. The paper was published in Pacific J. Math. **154** (1992), 103–129. In October 1997 a corrected and extended version was circulated; it added §2.5 on (G, M)-families, Proposition 3.7, §4.1, Proposition 4.3 and all of §5 (the 1997 footnotes recording this are merged into this paragraph). This 2026 version is built from a literal transcription of the 1997 text and its errata log (both in `original-1997/`).

What changed. Typos, misquoted normalizations and wrong references have been corrected silently; they are listed item by item in `CHANGES.md`. The substantive changes are:

1. the constant and the chamber in Macdonald's formula (Lemma 2.1) and the $\delta_P^{1/2}$ factor in Casselman's asymptotics (Lemma 3.3);
2. Lemma 3.5 is replaced by Brion's vertex decomposition (Lemma 3.5A);

3. Proposition 3.6 is restated as "vertex terms plus a T-independent remainder" and given a new, direct proof (the 1997 inductive argument did not control the T-independent remainder);
4. Proposition 4.2 and Theorem 4.5 are restated: $I(\chi, \phi)$ is defined as the T-independent part of the truncated kernel, and the spectral identity is proved under an explicit regularity hypothesis (R);
5. an example (§6.2, $SL(2)$) shows that (R) can fail and that the spectral side then needs atoms; the 1997 claim that $I(\chi, \gamma)$ is a rational function is corrected accordingly (Conjecture 4.6);
6. §5 is updated: the fundamental lemma is a theorem, and the 1997 closing question becomes Proposition 5.2.

New material: worked examples for $PGL(2)$ and $SL(2)$ (§6), the derivation from Arthur's local trace formula ("Route B", §7), a literature review before and after 1992 (§8) and a glossary (Appendix A).

Status.

Statement	Status (November 2026)
Truncated identity, Theorem 4.5(a)	proved
Vertex decomposition of J^T, I^T (Prop. 3.6, 4.2)	proved; details checked in rank one
Spectral identity under (R), Theorem 4.5(b)	proved
Hypothesis (R)	holds for $PGL(2)$ (§6.1); fails for $SL(2)$ (§6.2); open in rank ≥ 2
Atoms at non-regular characters, Conjecture 4.6	verified for $SL(2)$; open in general
κ-spectral measure for $SL(n)$, Prop. 5.1–5.2	conditional on Theorem 4.5 for H; checked for $SL(2)$
Route B (§7)	sketch; normalizations not checked

The numerical checks are `examples/pgl2/pgl2_check.py` and `examples/sl2/sl2_check.py` (run all with `python3 run_checks.py` at the top of the repository).

1 Introduction

Let G be a connected unramified reductive group over a p-adic field F, let G_{reg} denote the set of regular semisimple elements of $G(F)$, and let G_{ell} denote the set of regular elliptic elements of $G(F)$. Let $C_c^\infty(G(F))$ denote the algebra of locally constant compactly supported functions on $G(F)$ and let $\mathcal{H}(G, K)$ denote the commutative subalgebra of spherical (bi-K-invariant) functions, where K is a hyperspecial maximal compact subgroup of $G(F)$. Let $G(F)^1$ be the common kernel of the absolute values of the rational characters of $G(F)$ (§2.1); if G is semisimple, $G(F)^1 = G(F)$.

Let $\Phi \subset C_c^\infty(G(F))$ denote the subspace of functions supported in G_{ell}. Each $\phi \in \Phi$ has compactly generated support modulo conjugation in the sense of Kazhdan [Ka86, §4] (the 1997 text said "Ad-finite"). For each $\phi \in \Phi$, define

$$T_\phi : f \longmapsto \int_{G(F)^1 \times G(F)^1} f(h)\phi(g^{-1}hg)\, dg\, dh. \tag{1.1}$$

It is not hard to show that T_ϕ is an "elliptic" invariant distribution with compactly generated support (Lemma 4.4). In this note we restrict T_ϕ to $\mathcal{H}(G,K)$ and compute, in §§3–4, a (Macdonald–)Plancherel-type formula for T_ϕ:

$$T_\phi(f) = \int f^\vee(\chi)\, dI_\phi(\chi), \tag{1.2}$$

where χ runs over the compact torus $\widehat{T}^u$ of unitary unramified characters (modulo W) and dI_ϕ is a measure; see Theorem 4.5 below. For this, we use the geometric lemma of Bernstein and Zelevinsky, Macdonald's Plancherel formula, Macdonald's formula for the spherical function, results of Casselman on intertwining operators and asymptotics of the unramified principal series, and Brion's decomposition of lattice-point generating functions. This derivation follows the procedure of Waldspurger [Wa89a] rather closely, where the case of $GL(N)$ was worked out in detail. In §5, we examine briefly the "κ-stable" version of (1.1) for $SL(n)$:

$$T_\phi^\kappa : f \longmapsto I^\kappa(f,\phi) := \int_{SL(n,F)\times PGL(n,F)} \kappa(g) f(g^{-1}hg)\phi(h)\, dg\, dh, \tag{1.3}$$

where κ is an unramified character of $PGL(n,F)$ of order dividing n. After using the Weyl integration formula to obtain a "geometric" expansion for T_ϕ^κ, we use the fundamental lemma (now a theorem) to rewrite the result (see (5.6), (5.8) and Proposition 5.2 below).

Of course, the distribution T_ϕ also occurs in the context of Arthur's local trace formula [Ar89, Ar91]. Let R denote the unitary representation of $G(F)\times G(F)$ on $L^2(G(F))$ given by $(R(x_1,x_2)\psi)(y) := \psi(x_1^{-1}yx_2)$. Given $f=(f_1,f_2)$ in $C_c^\infty(G(F))\times C_c^\infty(G(F)) \hookrightarrow C_c^\infty(G(F)\times G(F))$, the kernel of the integral operator $R(f)$ is

$$K_f(x_1,x_2) = \int_{G(F)} f_1(x_1y) f_2(yx_2)\, dy = \int_{G(F)} f_1(y) f_2(x_1^{-1}yx_2)\, dy. \tag{1.4}$$

As in the global trace formula, one wants to find both a "geometric" and a "spectral" formula for a truncated version of the integral of $K_f(x,x)$. It should be emphasized that this is done below only for a very restricted class of $f=(f_1,f_2)$: f_1 spherical and f_2 supported on the elliptic set. For such f no truncation is needed on the geometric side (Lemma 4.4), and §7 explains how (1.2) follows from Arthur's formula.

Thus part of this paper could be viewed as a special case of Arthur's local trace formula or as a generalization of part of Waldspurger's work [Wa89a]. Another way one might interpret the distributions $\phi \mapsto I(\chi,\phi)$ is as follows. We will see in §4 below (Proposition 4.2) that $\phi \longmapsto I(\chi,\phi)$ is G-admissible in the sense of [HC78]. Then, regarding this invariant distribution as a function (the existence of which is assured by [HC78, Theorem 19]), we may rewrite (1.2) as

$$\int_{G(F)^1} f(g^{-1}\gamma g)\, dg = \int f^\vee(\chi)\, dI_\gamma(\chi), \tag{1.5}$$

for $\gamma \in G_{\mathrm{ell}}$ and f spherical. Thus (1.5) gives a relatively explicit formula for the unramified part of the "spectral support" of an elliptic orbital integral on $G(F)$. (This was to be the subject of [Jo97], announced in 1997 as "in preparation" but never written up.)

In the 1997 version it was asserted that $dI_\gamma = I(\chi,\gamma)\, d\mu(\chi)$ with $I(\cdot,\gamma)$ a rational function on the complex torus of unramified characters (whose compact real form carries the Plancherel measure, [Ma71]). This is correct for $PGL(2)$ (§6.1), where $I(\cdot,\gamma)$ has double poles at the non-regular characters that are cancelled by the double zeros of the Plancherel density. It is *not*

correct for $SL(2)$ (§6.2): there dI_γ has, besides an absolutely continuous part, a point mass at the unramified quadratic character χ_0. This is a non-regular unitary character at which the Plancherel density does not vanish, and $i_{G,A}(\chi_0)$ is reducible. The atom is half the κ-orbital integral, $\frac{1}{2}f^\vee(\chi_0)$, as the fundamental lemma predicts. Theorem 4.5 and Conjecture 4.6 state the corrected form.

A somewhat analogous formula to (1.5), for stable unipotent orbits, was conjectured in [As91]. Assem's conjecture is a theorem for $GL(n)$, and for $GL(n)$ the germ expansion and Assem's formula yield a relatively explicit formula for $I(\chi,\gamma)$ (this idea can essentially be found in [Wa89a]). Finally, in the case of $SL(n)$ the fundamental lemma of Waldspurger [Wa91] may be reformulated as a functorial property of the spectral measures dI^κ_γ (Proposition 5.2).

Road map. The proof of (1.2) has three steps.

1. *Truncated inner products* (§3). For a truncation parameter T, the integral J^T of a product of two matrix coefficients over $G(T) = \bigcup_a KaK$ (the finitely many double cosets "below T") is computed from Casselman's asymptotics on the regions where they apply. Brion's theorem turns the resulting lattice-point sums into vertex terms $\psi(T_Q)R_Q$, one for each standard parabolic Q; the vertex $T_G = 0$ gives a T-independent term (Proposition 3.6).
2. *The truncated kernel* (§4.2). Expanding the spherical function in matrix coefficients expresses $I^T(\chi,\phi) = \int_{G(T)\times G} \Gamma_\chi(g^{-1}hg)\phi(h)$ through J^T. So I^T is a sum of vertex terms, and its T-independent part defines $I(\chi,\phi)$ (Proposition 4.2).
3. *Removing the truncation* (§4.3). For T large, $T_\phi(f) = \int f^\vee I^T\,d\mu$ exactly (Fubini on a compact set). If the vertex terms are regular enough (Hypothesis (R)), their contributions vanish as $T\to\infty$ and $T_\phi(f) = \int f^\vee I\,d\mu$ (Theorem 4.5). Where (R) fails they leave atoms (Conjecture 4.6, §6.2).

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2 Notation and background

2.0 Conventions (new)

These conventions are fixed once and for all; the 1997 text left several of them implicit, and some of its formulas mixed conventions.

- F is a p-adic field with ring of integers $\mathcal{O}$, uniformizer π, residue field of order q, $|\pi|_F = q^{-1}$, and normalized valuation ν_F.
- A is a maximal F-split torus, $M_0 = Z_G(A)$ its centralizer (a maximal torus, since G is quasi-split), $B = M_0N_0 \supset M_0$ a Borel subgroup over F, and K a hyperspecial maximal compact subgroup in good position relative to A. $I \subset K$ is the Iwahori subgroup attached to B.
- *Roots and weights.* Δ is the set of simple (reduced) roots of A in B, $\Delta^\vee = \{\alpha^\vee\}$ the simple coroots, $\hat\Delta = \{\varpi_\alpha\}$ the fundamental weights (in $(\mathfrak{a}_0^G)^*$), so that $\langle\alpha^\vee,\varpi_\beta\rangle = \delta_{\alpha\beta}$. This is Arthur's notation; the 1997 text used $\Delta^\vee$, $\hat\alpha$, $\alpha^\wedge$ and $R^\wedge$ for overlapping objects.

- $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$ (written $\mathfrak{a}_B$ or $\mathcal{A}_{A,\mathbb{R}}$ in 1997). For a standard parabolic $P = MN$, $\mathfrak{a}_0 = \mathfrak{a}_P \oplus \mathfrak{a}_0^P$, and we write $X = X_P + X^P$. (The 1997 text wrote X_M, X^M, with the two subscripts interchanged in one place.)
- *Chamber.* $A^- = \{a \in A(F) : |\alpha(a)|_F \leq 1 \ \forall \alpha \in \Delta\}$. Since $|\alpha(a)|_F = q^{-\langle \nu_A(a), \alpha \rangle}$ (see (2.4)), $\nu_A(A^-)$ is the closed dominant cone $X_*(A)^+$. Elements of A^- contract N_0. For a standard parabolic $P = MN$ and $0 < \epsilon \leq 1$ put

$$A_P^-(\epsilon) = \{a \in A^- : \ |\alpha(a)|_F \leq \epsilon \ \ \forall \alpha \in \Delta \smallsetminus \Delta^M\}, \qquad A^-(\epsilon) = A_B^-(\epsilon).$$

- *Induction and Jacquet modules* are normalized: $i_{G,M}(\sigma) = \{f : G \to V_\sigma : \ f(mng) = \delta_P^{1/2}(m)\sigma(m)f(g)\}$ with G acting by right translation, and $r_{M,G}(\pi) = \delta_P^{-1/2} \otimes \pi_N$. Here $\delta_P(m) = |\det \mathrm{Ad}(m)|_{\mathrm{Lie}\, N}|_F$, so $\delta_B(a) = q^{-\langle \nu_A(a), 2\rho \rangle} \leq 1$ on A^-.
- *Weyl group.* $W = N_G(A)/M_0$, with representatives in K; $(w\chi)(a) = \chi(w^{-1}aw)$. For a standard Levi M, $[W_M \backslash W]$ denotes the set of minimal-length representatives w of the cosets $W_M w$ (equivalently $w^{-1}\Delta^M > 0$). All sums "over W/W_M" in the 1997 text should be read as sums over $[W_M \backslash W]$.
- *Unramified characters.* For $\lambda \in \mathfrak{a}_{0,\mathbb{C}}^*$ put $\chi_\lambda(a) = q^{-\langle \nu_A(a), \lambda \rangle}$. This is the one definition of χ_λ used throughout. (The 1997 text gave three different, ill-typed definitions.)
- *Measures.* $\mathrm{vol}(K) = \mathrm{vol}(K \cap A(F)) = \mathrm{vol}(K \cap N(F)) = 1$, and $\mathrm{vol}(K \cap M(F)) = 1$ for every standard Levi M. The Haar measure on $\widehat{T}^u$ (below) has total mass 1.
- *Symbols.* T always denotes the truncation parameter. Maximal tori are M_0 or (for elliptic tori) S. Characteristic functions are $\mathbf{1}_E$. Q_G is Macdonald's constant. The letters P, Q denote parabolic subgroups. We avoid $\chi(M)$ for a characteristic function.

2.1 Root spaces

Let G be a connected unramified reductive group of semi-simple rank ℓ over F, split over a finite unramified extension E/F. (Recall that a reductive group G over F is unramified if it is quasi-split over F and split over a finite unramified extension [Ca79, p. 135].) Let $M_0 = Z_G(A)$, B and A be as in §2.0. Let $\Gamma = \mathrm{Gal}(E/F)$, let $X^* = X^*(M_0)$ denote the character lattice of M_0, and let $\Sigma_{M_0} \subset X^*$ denote the (absolute) root system. The Γ-module structure of X^* leaves Σ_{M_0} invariant. Let Σ denote the set of reduced roots of G relative to A and Δ the corresponding reduced fundamental system. The character lattice

$$X^*(A) = \mathrm{Hom}_{F\text{-grps}}(A, GL(1)) \tag{2.1}$$

may be regarded as a quotient of X^* containing Σ. Let

$$X_*(A) := \mathrm{Hom}_{\mathbb{Z}}(X^*(A), \mathbb{Z}) \tag{2.2}$$

denote the co-character lattice. Let $\Sigma^+ \subset \Sigma$ be the set of positive roots containing Δ. For parabolic subgroups $P \subset Q$ containing A, let Δ_P^Q denote the set of simple roots of $(P \cap M_Q, A_P)$, where $Q = M_Q N_Q$ is the Levi decomposition and A_P is the split component of the center of M_P [Ar81]; $\Delta_P^G = \Delta_P$. Each $w \in W$ has a representative in K via $N_G(A)/Z_G(A) \cong (K \cap N_G(A))/(K \cap Z_G(A))$ [Ca79, p. 140]. W is generated by the simple reflections $\{w_\alpha : \alpha \in \Delta\}$ ([Cas95], §1). Let R denote the root lattice generated by Δ and L the coroot lattice generated by

$\Delta^\vee$. If G is semi-simple these lattices and $X^*(A)$, $X_*(A)$ all have rank ℓ, and $X^*(A)/R$, $X_*(A)/L$ are finite. If G is split, G is adjoint iff $X^*(A) = R$ and simply connected iff $X_*(A) = L$; for non-split G these criteria must be phrased with the absolute root datum (quasi-split $SU(3)$ and $PU(3)$ both have $X^*(A) = R$).

For each $\theta \subset \Delta$, let $P_\theta = M_\theta N_\theta \supset B$ be the corresponding standard parabolic, so $P_\emptyset = B$ and $M_\emptyset = M_0$; let $A_M = A_\theta$ be the split component of the center of $M = M_\theta$. A Levi component $M = M_P$ of a parabolic $P = M_P N_P$ is called a Levi subgroup of G; all Levi and parabolic subgroups are taken to contain A. $\mathcal{P}(M)$ is the set of parabolics with Levi component M. For $P \subset Q$ there are orthogonal decompositions $\mathfrak{a}_P = \mathfrak{a}_Q \oplus \mathfrak{a}_P^Q$ and $\mathfrak{a}_P^* = \mathfrak{a}_Q^* \oplus (\mathfrak{a}_P^Q)^*$ [Ar78]. The projections of $(\Delta_P^Q)^\vee$ form a basis of $\mathfrak{a}_P^Q$ and $\hat{\Delta}_P^Q$ a basis of $(\mathfrak{a}_P^Q)^*$. Let τ_P^Q and $\hat{\tau}_P^Q$ denote the characteristic functions on $\mathfrak{a}_0$ of

$$\{X:\ \alpha(X) > 0,\ \alpha \in \Delta_P^Q\} \qquad \text{and} \qquad \{X:\ \varpi(X) > 0,\ \varpi \in \hat{\Delta}_P^Q\},$$

respectively, as in [Ar78].

The kernel of the map $H_M : M(F) \to \mathfrak{a}_P$ of [Ar78] is denoted $M(F)^1$; it is the intersection of the kernels of the absolute values of the rational characters of $M(F)$. The Haar measure on $M(F)^1$ is the one determined by those on $M(F)$ and $\mathfrak{a}_P$.

The pairing $\langle \cdot, \cdot \rangle : X_*(A) \times X^*(A) \to \mathbb{Z}$, $\langle \lambda_*, \lambda^* \rangle := \lambda_*(\lambda^*)$, identifies $X^*(A)$ with the dual lattice of $X_*(A)$. It gives an isomorphism

$$\mathrm{Hom}_{F\text{-grps}}(GL(1), A) \cong X_*(A), \tag{2.3}$$

associating to λ the unique λ_* with $\lambda^*(\lambda(t)) = t^{\langle \lambda_*, \lambda^* \rangle}$ for all $t \in F^\times$, $\lambda^* \in X^*(A)$. Let $a_\alpha \in A(F)$ denote $\alpha^\vee(\pi)$ (for non-split G, the element of [Ca79, §3] used in Casselman's formula (2.8)). If G is split then: (a) $\delta_B(a_\alpha) = q^{-2}$ for $\alpha \in \Delta$; (b) the a_α generate the image of the coroot lattice L in $A(F)/(A(F) \cap K)$, freely, and this image is everything iff G is simply connected (in general it has finite index $[X_*(A) : L]$); (c) $w_\alpha^{-1} a w_\alpha = a\, a_\alpha^{-\langle \nu_A(a), \alpha \rangle}$ modulo $A(F) \cap K$, where ν_A is as in (2.4) ([Ca79], pp. 141–142, [Ma71], pp. 42–43). To each $\beta \in \Sigma \cup \frac{1}{2}\Sigma$ there is attached a number $q_\beta > 0$ as in [Ca79]; if G is split and $\alpha \in \Sigma$ then $q_\alpha = q$ and $q_{\alpha/2} = 1$.

Let $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$ and $\mathfrak{a}_0^* = X^*(A) \otimes \mathbb{R}$, and extend $\langle \cdot, \cdot \rangle$ to $\mathfrak{a}_0 \times \mathfrak{a}_0^*$. Then $\Delta^\vee \subset \mathfrak{a}_0$ and $\hat{\Delta} \subset \mathfrak{a}_0^*$ are dual bases (for semisimple G): $\langle \alpha^\vee, \varpi_\beta \rangle = \delta_{\alpha\beta}$ for $\alpha, \beta \in \Delta$. (The 1997 text wrote $\langle \alpha^\vee, \beta \rangle = 2\delta_{\alpha\beta}$; the left side is the Cartan matrix.)

There is a surjection $\nu_A : A(F) \to X_*(A)$ characterized by

$$\langle \nu_A(a), \lambda^* \rangle = \nu_F(\lambda^*(a)), \qquad \forall \lambda^* \in X^*(A),\ a \in A(F). \tag{2.4}$$

Thus we obtain an isomorphism

$$\nu_A^{-1} : X_*(A) \xrightarrow{\sim} A(F)/(A(F) \cap K) \ \cong\ M_0(F)/(M_0(F) \cap K), \tag{2.5}$$

and every unramified character of $M_0(F)$ may be identified with a character of the discrete group $X_*(A)$. We write $a_\lambda = \nu_A^{-1}(\lambda)$ and $\chi(\lambda) = \chi(a_\lambda)$. For each Levi M, $X_*(A_M) = X_*(A) \cap \mathfrak{a}_M$.

Tori of characters. Let $L^\vee = \frac{2\pi i}{\log q} X^*(A) \subset \mathfrak{a}_{0,\mathbb{C}}^*$. The map $\lambda \mapsto \chi_\lambda$ gives isomorphisms

$$\widehat{T} := \mathrm{Hom}(X_*(A), \mathbb{C}^\times) \cong \mathfrak{a}_{0,\mathbb{C}}^*/L^\vee, \qquad \widehat{T}^u := \mathrm{Hom}(X_*(A), \mathbb{C}^{\times 1}) \cong i\mathfrak{a}_0^*/L^\vee, \tag{2.6}$$

where $\mathbb{C}^{\times 1} = \{z : |z| = 1\}$. Here $\widehat{T}$ is the complex torus of all unramified characters and $\widehat{T}^u$, the Pontryagin dual of $X_*(A)$, is its compact real form: the unitary unramified characters. Thus

$$\mathrm{Hom}_{unr}(A(F), \mathbb{C}^{\times 1}) \cong i\mathfrak{a}_0^*/L^\vee, \tag{2.7}$$

where $\mathrm{Hom}_{unr}(H(F), \mathrm{End}\, V)$ denotes the continuous homomorphisms with a non-zero $H(F)\cap K$-fixed vector. (The 1997 text used the lattice $\{\lambda : \lambda(\hat{\alpha}) \in \mathbb{Z}\}$, which is correct only for adjoint G and has the wrong scaling, and conflated $\widehat{T}$ with $\widehat{T}^u$.)

2.2 Intertwining operators

For the unramified principal series representations $i_{G,A}(\chi) = V_B(\chi)$ of $G(F)$ attached to an unramified character χ of $M_0(F)$ we refer to [Ca79]. The pairing $\langle\cdot,\cdot\rangle_G$ on $V_B(\chi)\times V_B(\chi^{-1})$, $\langle f, f'\rangle_G = \int_K f(k)f'(k)\,dk$, identifies the contragredient of $i_{G,A}(\chi)$ with $i_{G,A}(\chi^{-1})$.

Let χ be a regular unramified character (so the $w\chi$, $w \in W$, are distinct), and let $T_w : V_B(\chi) \to V_B(w\chi)$ denote the intertwining operator of [Ca79]. If $\Phi_{K,\chi} \in V_B(\chi)^K$ is the K-fixed vector with $\Phi_{K,\chi}(1) = 1$ then Casselman [Cas80, Thm. 3.1], [Ca79, Thm. 3.9] has shown that

$$T_w(\Phi_{K,\chi}) = c_w(\chi)\Phi_{K,w\chi}, \qquad c_w(\chi) = \prod_{\alpha\in\Sigma^+,\ w\alpha<0} c_\alpha(\chi), \tag{2.8}$$

$$c_\alpha(\chi) = \frac{\big(1 - q_{\alpha/2}^{-1/2} q_\alpha^{-1}\chi(a_\alpha)\big)\big(1 + q_{\alpha/2}^{-1/2}\chi(a_\alpha)\big)}{1 - \chi(a_\alpha)^2}.$$

In the split case $c_\alpha(\chi) = (1 - q^{-1}\chi(a_\alpha))/(1 - \chi(a_\alpha))$. Put $c(\chi) = c_{w_\ell}(\chi) = \prod_{\alpha>0} c_\alpha(\chi)$, w_ℓ the longest element of W. Also $T_{w_1w_2} = T_{w_1}T_{w_2}$ if $\ell(w_1w_2) = \ell(w_1) + \ell(w_2)$; here $\ell(w)$ is the length of w, which is also the number of walls in $\mathfrak{a}_0$ separating the positive chambers of B^w and B [Cas95, ch. 1].

Remark 2026.1 (10 Nov. 2026) (exponent in c_α). The 1997 text printed $q_{\alpha/2}^{+1/2}$ in the first factor of c_α; we use Casselman's $q_{\alpha/2}^{-1/2}$. The two agree for split G. The non-split formula has not been re-derived here; all numerical checks in this paper are for split groups.

2.3 Plancherel's formula and Macdonald's formula

Let $\mathcal{H}(G,K)$ denote the algebra of bi-K-invariant functions in $C_c^\infty(G(F))$ and $C^\infty(G,K)$ the analogous space of locally constant bi-K-invariant functions. For $f \in V_B(\chi)$ define $f^\natural(g) = \int_K f(kg)\,dk$, and let $\Gamma_\chi = \Phi_{K,\chi}^\natural$ be the zonal spherical function; $\Gamma_\chi(g) = \langle i_{G,A}(\chi)(g)\Phi_{K,\chi}, \Phi_{K,\chi^{-1}}\rangle_G$. Macdonald's formula states:

Lemma 2.1 (Macdonald)**.** *If χ is an unramified regular character of $M_0(F)$ then, for $a \in A^-$,*

$$\Gamma_\chi(a) = Q_G^{-1}\,\delta_B(a)^{1/2} \sum_{w\in W} c\big((w\chi)^{-1}\big)\,(w\chi)(a), \qquad Q_G := \sum_{w\in W} (IwI : I)^{-1},$$

which is $\sum_w q^{-\ell(w)}$ in the split case.

Remark 2026.2 (10 Nov. 2026) (Lemma 2.1). Two corrections to the 1997 statement, which read $\Gamma_\chi(a) = Q^{-1}\delta_B^{1/2}(a)\sum_w c(w\chi)(w\chi)(a)$ for all $a \in A(F)$ with $Q = \sum_w (IwI : I)$.

1. The constant is $\sum_w (IwI : I)^{-1}$; for $PGL(2)$ and $SL(2)$, $\Gamma_\chi(1) = 1$ forces $Q_G = 1 + q^{-1}$, and the Plancherel inversion (Corollary 2.3) holds with $1+q^{-1}$ and fails with $1+q$ (checked numerically, §6).

2. The formula holds only on the closed chamber A^- (the right side is not W-invariant in a), and on the chamber A^- defined in §2.0 the coefficient of $(w\chi)(a)$ is $c((w\chi)^{-1})$; the 1997 coefficient $c(w\chi)$ belongs to the opposite chamber $A^+ = (A^-)^{-1}$. Verified to machine precision for $PGL(2)$ and $SL(2)$. In general the pairing "chamber versus c-function" depends on the convention for T_w; we have checked it only in rank one.

For $f \in \mathcal{H}(G, K)$ and χ unramified, define the Fourier (Satake) transform of f at χ by

$$f^\vee(\chi) := \int_{G(F)^1} f(g)\Gamma_\chi(g^{-1})\, dg. \tag{2.9}$$

Let $\Omega_K(G)$ denote the set of zonal spherical functions of $G(F)^1$ relative to K:

$$\Omega_K(G) := \{\omega \in C^\infty(G, K) :\ \omega(1) = 1 \text{ and } f * \omega = \lambda_f \omega \text{ for all } f \in \mathcal{H}(G, K),\ \text{some } \lambda_f \in \mathbb{C}\}. \tag{2.10}$$

Let $\Omega_K^+(G)$ denote the subset of positive definite zonal spherical functions. If χ is unitary then $\Gamma_\chi \in \Omega_K^+(G)$ [Ma71, Thm. 3.3.12]. The Fourier transform of $f \in \mathcal{H}(G, K)$ at $\omega \in \Omega_K^+(G)$ is

$$f^\vee(\omega) := \int_{G(F)^1} f(g)\omega(g^{-1})\, dg. \tag{2.11}$$

The relation between the Fourier transform and the Satake transform is given in [Ma71, p. 47]. The Plancherel measure $d\mu(\omega)$ is the positive measure on $\Omega_K^+(G)$ such that, for all $f \in \mathcal{H}(G, K)$,

$$f^\vee(\omega) \in L^2(\Omega_K^+(G), d\mu) \tag{2.12}$$

and

$$\int_{G(F)^1} |f(g)|^2\, dg = \int_{\Omega_K^+(G)} |f^\vee(\omega)|^2\, d\mu(\omega). \tag{2.13}$$

By a theorem of Godement, such a measure exists and is unique.

Lemma 2.2 (Macdonald [Ma71, Thm. 5.1.2])**.** *The support of the Plancherel measure is the image of the compact torus $\widehat{T}^u$ of (2.6) (modulo W). Let $d\chi$ be the Haar probability measure on $\widehat{T}^u \cong i\mathfrak{a}_0^*/L^\vee$. The Plancherel measure of $G(F)$ with respect to K is*

$$d\mu(\chi) = \frac{Q_G}{|W|}\, |c(\chi)|^{-2}\, d\chi.$$

Corollary 2.3. *For all $f \in \mathcal{H}(G, K)$,*

$$f(g) = \int_{\Omega_K^+(G)} f^\vee(\omega)\overline{\omega(g^{-1})}\, d\mu(\omega) = \int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g)\, d\mu(\chi).$$

For all $f_1, f_2 \in \mathcal{H}(G, K)$,

$$\int_{G(F)^1} f_1(g)\overline{f_2(g)}\, dg = \int_{\Omega_K^+(G)} f_1^\vee(\omega)\overline{f_2^\vee(\omega)}\, d\mu(\omega) = \int_{\widehat{T}^u} f_1^\vee(\chi)\overline{f_2^\vee(\chi)}\, d\mu(\chi).$$

Proof. First, $d\mu$ is supported on $\widehat{T}^u/W$ [Ma71, Thm. 5.1.2], every ω is a Γ_χ [Ma71, Thm. 3.3.12], $\Gamma_\chi(g^{-1}) = \Gamma_{\chi^{-1}}(g)$ [Ma71, Prop. 3.3.2], and $\overline{\Gamma_{\chi^{-1}}(g)} = \Gamma_\chi(g)$ for unitary χ [Ca79, p. 150]. So in each statement the second equality follows from the first. The second statement follows from (2.13) by polarization: replace f by $f_1 + f_2$ and by $f_1 + if_2$ in (2.13), expand, and combine; the first gives the real part and the second the imaginary part of the cross term. The first statement follows from the second by taking f_2 to be the characteristic function of a double coset KgK and using the definition (2.11). □

2.4 The Jacquet functor

For any parabolic $P = MN$ of G we have $G(F) = P(F)K$, and for each Levi M and parabolic P^M of M we have $M(F) = P^M(F)(K \cap M(F))$ [Ca79, p. 140]. In this case the notion of "compactly induced" representations [BZ77, §1.8] agrees with the usual notion of unitarily induced representations. Let

$$i_{G,M} : \operatorname{Alg} M \to \operatorname{Alg} G \tag{2.14}$$

denote normalized induction, in the notation of [BZ77] (so $i_{G,M}(\tau) = V_P(\tau)$ in Arthur's notation), and let

$$r_{M,G} : \operatorname{Alg} G \to \operatorname{Alg} M \tag{2.15}$$

denote the normalized Jacquet functor [BZ77, §2.3] (called the first Jacquet functor in [Ca79, §2.2]). We sometimes write $\pi_N = r_{M,G}(\pi)$.

Let W_M denote the Weyl group of M. The special case of the geometric lemma of Bernstein–Zelevinsky which we need is the following.

Lemma 2.4 ([BZ77, §2.12)**.** *] There is an enumeration $w_1, \dots, w_k$ of $[W_M \backslash W]$ such that, for each character χ of $M_0(F)$, the M-module $r_{M,G} \circ i_{G,A}(\chi)$ has a filtration*

$$r_{M,G} \circ i_{G,A}(\chi) = V_k \supset V_{k-1} \supset \cdots \supset V_1 \supset V_0 = \{0\} \qquad \textit{with} \qquad V_j/V_{j-1} \cong i_{M,A}(w_j\chi).$$

The subquotients need not be irreducible (take $M = G$). Let $V_{(s)}$ denote the semi-simplification of an M-module V.

Corollary 2.5. *For each character χ of $M_0(F)$, $\big(r_{M,G} \circ i_{G,A}(\chi)\big)_{(s)} \cong \bigoplus_{w \in [W_M \backslash W]} i_{M,A}(w\chi)_{(s)}$.*

Proof. Follows from Lemma 2.4 and the definition of the semi-simplification. □

Remark 1. If χ is regular, then $r_{M,G} \circ i_{G,A}(\chi) \cong \bigoplus_{w \in [W_M \backslash W]} i_{M,A}(w\chi)$, a direct sum of (possibly reducible) M-modules. Indeed the cuspidal supports $W_M \cdot w\chi$ of the summands are distinct for distinct cosets $W_M w$ (if $w'\chi = vw\chi$ with $v \in W_M$ then $w' = vw$ by regularity), and a finite-length M-module splits according to the generalized eigencharacters of the Bernstein center of M, which separate cuspidal supports. The 1997 text said instead that $r_{M,G} \circ i_{G,A}(\chi)$ is semisimple; that is false for $M = G$ (for $\chi = \delta_B^{1/2}$ on $SL(2)$, $i_{G,A}(\chi)$ is a non-split extension of the trivial representation by the Steinberg representation).

2.5 (G, M)-families

These results are used in §4 only to explain where the regularity Hypothesis (R) comes from; see Remark 2026.9. (In the 1997 text this subsection was said to be "not needed".)

For a collection $\{c_P(\lambda) : P \in \mathcal{P}(M)\}$ of smooth functions on $i\mathfrak{a}_M^*$ we say that it is a (G,M)-**family** if for any adjacent $P, P' \in \mathcal{P}(M)$, $c_P(\lambda) = c_{P'}(\lambda)$ on the hyperplane spanned by the common wall of the chambers of P and P' [Ar81, §6]. For a collection $\{\tilde{c}_P(\chi) : P \in \mathcal{P}(A)\}$ of smooth functions on $\widehat{T}^u$ we say that it is a $(G,A)^\sim$**-family** if for any adjacent $P, P' \in \mathcal{P}(A)$ we have $\tilde{c}_P(\chi_\lambda) = \tilde{c}_{P'}(\chi_\lambda)$ on the set

$$\Big\{\lambda \in i\mathfrak{a}_0^* : \ \langle \alpha^\vee, \lambda\rangle \in \frac{2\pi i}{\log q}\mathbb{Z}\Big\} = \{\lambda : \ \chi_\lambda(a_\alpha) = 1\},$$

where α is one of the two roots separating P and P' [Wa89a, p. 26].

Let $Q \subset R$ be parabolic subgroups and let L_Q^R denote the lattice generated by the projections of $\{\alpha^\vee : \alpha \in \Delta_Q^R\}$. Let

$$\theta_Q^R(\lambda) = \mathrm{vol}(\mathfrak{a}_Q^R/L_Q^R)^{-1} \prod_{\alpha \in \Delta_Q^R} \lambda(\alpha^\vee), \qquad \lambda \in i\mathfrak{a}_Q^*, \tag{2.16}$$

and define $\tilde{\theta}_B = \tilde{\theta}$ by

$$\tilde{\theta}(\chi) := \prod_{\alpha \in \Delta} (1 - \chi(a_\alpha)). \tag{2.17}$$

Lemma 2.6 ([Ar81, Lemma 6.2)**.** *] If $\{c_P(\lambda) : P \in \mathcal{P}(M)\}$ is a (G,M)-family then $c_M(\lambda) := \sum_{P \in \mathcal{P}(M)} c_P(\lambda)\theta_P(\lambda)^{-1}$ extends to a smooth function on $i\mathfrak{a}_M^*$.*

Lemma 2.7 ([Wa89a, §7)**.** *] If $\{\tilde{c}_P(\chi) : P \in \mathcal{P}(A)\}$ is a $(G,A)^\sim$-family then*

$$\tilde{c}_A(\chi) := \sum_{P \in \mathcal{P}(A)} \tilde{c}_P(\chi)\tilde{\theta}_P(\chi)^{-1}$$

extends to a smooth function on $\widehat{T}^u$. Here $\tilde{\theta}_P(\chi) := \tilde{\theta}_B(w_P\chi)$, where $\tilde{\theta}_B$ is defined in (2.17) and $w_P \in W$ is the unique element with $P = w_P^{-1}Bw_P$.

The factor $\tilde{\theta}$ is the periodic analogue of θ:

Lemma 2.8 ([Wa89a, II.7.1)**.** *] There exists $u_B \in C_c^\infty(i\mathfrak{a}_0^*/i\mathfrak{a}_G^*)$ such that*

(a) for all $\lambda \in i\mathfrak{a}_0^*$, $\sum_{r \in L^\vee} u_B(\lambda + r)\,\theta_B(\lambda + r)^{-1} = \tilde{\theta}_B(\chi_\lambda)^{-1}$;

(b) if $r \in L^\vee$ lies in the support of u_B then $r = 0$.

(One may take $u_B = \eta\,\theta_B/\tilde{\theta}_B(\chi_\cdot)$, where $\eta \in C_c^\infty$ is supported in a small neighborhood of 0 and $\sum_{r \in L^\vee} \eta(\lambda + r) = 1$; then u_B is smooth because $\theta_B(\lambda)/\tilde{\theta}_B(\chi_\lambda)$ is smooth near 0, and (a) holds by periodicity of $\tilde{\theta}_B(\chi_\lambda)$.)

Lemma 2.9 ([Wa89a, §7)**.** *] Let $\{\tilde{c}_P(\chi) : P \in \mathcal{P}(A)\}$ be a $(G,A)^\sim$-family. For $P \in \mathcal{P}(A)$ let $c_P(\lambda) = \tilde{c}_P(\chi_\lambda)u_P(\lambda)$, where $u_P(\lambda) = u_B(w_P\lambda)$ and u_B is as in Lemma 2.8. Then $\{c_P\}_P$ is a (G,A)-family, and*

$$\tilde{c}_A(\chi_\lambda) = \sum_{r \in L^\vee} c_A(\lambda + r), \qquad c_A(\lambda) = \sum_P c_P(\lambda)\theta_P(\lambda)^{-1}.$$

(The 1997 statement attached an index r to the family and left a dangling P on the right; since $\chi_{\lambda+r} = \chi_\lambda$ for $r \in L^\vee$, the version above is what the proof via Lemma 2.8 gives.)

Remark 2. Lemma 2.9 follows from Lemmas 2.6–2.8. These results relate the $(G,A)^\sim$-families occurring implicitly in §4 to Arthur's (G,A)-families [Ar81].

Remark 2026.3 (10 Nov. 2026) (§2.5). Lemmas 2.6 and 2.8 are standard (2.8 by the construction given). The precise form of Lemmas 2.7 and 2.9 has not been re-checked against [Wa89a, §7]; they are not used in any proof below.

3 Inner products of some matrix coefficients

3.1 Matrix coefficients

Assume as before that G is connected, unramified and reductive, and use the measures of §2.0. Let χ be an unramified regular character of $M_0(F)$ and $V_B(\chi) = i_{G,A}(\chi)$. For $n \geq 1$ let $K_n \subset K$ be the n-th principal congruence subgroup; it is normal in K and has an Iwahori factorization with respect to every standard parabolic. A vector has *level* n if it is fixed by K_n. Write $\pi = \pi_\chi = i_{G,A}(\chi)$; its restriction to K does not depend on χ (compact picture).

Casselman's asymptotics. Let $P = MN$ be standard and $\bar{P} = M\bar{N}$ its opposite. By Casselman [Cas95, §4, Prop. 4.3.3] there is a canonical M-invariant pairing $\langle\cdot,\cdot\rangle_N$ between $r_{M,G}(\pi_\chi)$ (taken with respect to P) and $r_{M,\bar{P}}(\pi_{\chi^{-1}})$ (taken with respect to $\bar{P}$), and for every n there is $\epsilon = \epsilon(n) \in (0,1]$, independent of the unramified character χ, such that for all $f \in V_B(\chi)$, $f' \in V_B(\chi^{-1})$ of level n

$$\langle \pi_\chi(a)f, f'\rangle_G = \delta_P^{1/2}(a)\, \langle r_{M,G}(\pi_\chi)(a)f_N,\, f'_{\bar{N}}\rangle_N \qquad \text{for all } a \in A_P^-(\epsilon). \tag{3.1}$$

For $P = B$ this is the case $a \in A^-(\epsilon)$. The independence of ϵ from χ follows from [Ca79, §3].

Remark 2026.4 (10 Nov. 2026) ((3.1)). Three corrections to the 1997 version of (3.1): the second space is the Jacquet module for the *opposite* parabolic; the factor $\delta_P^{1/2}(a)$ was missing; and ϵ depends on the level of f, f'. We use (3.1) on $A_P^-(\epsilon)$, i.e. for a deep only in the N-directions. This is how Casselman's argument runs (it uses only that a contracts N_0 and contracts $N \cap K_n$ strongly), but we have not checked that [Cas95, Prop. 4.3.3] is stated in exactly this generality rather than for a in the split center of M.

For $f \in V_B(w\chi)$ let $R_M f = \delta_P^{-1/2} \cdot f|_{M(F)} \in i_{M,A}(w\chi)$; the map R_M factors through $r_{M,G}$ and is M-equivariant [Wa89a, §I]. For $w \in [W_M\backslash W]$ the maps $R_M \circ T_w$ give the decomposition of Remark 1,

$$r_{M,G}(\pi_\chi) \xrightarrow{\sim} \bigoplus_{w\in[W_M\backslash W]} i_{M,A}(w\chi), \qquad f_N \longmapsto (f_w)_w, \quad f_w := R_M T_w f. \tag{3.2}$$

Lemma 3.1. *For χ unramified regular, $f \in V_B(\chi)$ and $a \in A(F)$, $\big(\pi_\chi(a)f\big)_w = i_{M,A}(w\chi)(a) f_w$ for every $w \in [W_M\backslash W]$.*

Proof. R_M and T_w are intertwining maps, so $R_M T_w \pi_\chi(a) f = i_{M,A}(w\chi)(a) R_M T_w f$. □

Lemma 3.2. *Let χ be regular. In terms of the decomposition (3.2) of $r_{M,G}(\pi_\chi)$ and the corresponding decomposition of $r_{M,\bar{P}}(\pi_{\chi^{-1}})$ into summands isomorphic to $i_{M,A}((w\chi)^{-1})$, $w \in [W_M\backslash W]$, the pairing $\langle\cdot,\cdot\rangle_N$ is block-diagonal, and on the w-th block it is a scalar multiple of the standard pairing*

$$\langle u, u'\rangle_M := \int_{M(F)\cap K} u(k)u'(k)\,dk$$

between $i_{M,A}(w\chi)$ and $i_{M,A}((w\chi)^{-1})$.

Remark 3. For our choice of Haar measures, if u, u' are $(M\cap K)$-fixed vectors in $i_{M,A}(w\chi)$ and $i_{M,A}((w\chi)^{-1})$, then $\langle u, u'\rangle_M = u(1)u'(1)$.

Proof of Lemma 3.2. The pairing $\langle\cdot,\cdot\rangle_N$ is M-invariant, so its restriction to the (w, w') block lies in $\mathrm{Hom}_M\left(i_{M,A}(w\chi)\otimes i_{M,A}((w'\chi)^{-1}), \mathbb{C}\right) = \mathrm{Hom}_M\left(i_{M,A}(w\chi), i_{M,A}(w'\chi)\right)$. A non-zero element forces a common cuspidal support, $W_M w\chi = W_M w'\chi$, hence $W_M w = W_M w'$ by regularity. For $w = w'$, Bruhat theory gives $\dim \mathrm{End}_M(i_{M,A}(w\chi)) \le \#\{v \in W_M : vw\chi = w\chi\} = 1$. □

(The 1997 proof argued that the pairing is "of the form $\sum a_{w,w'}\langle z_w, z'_{w'}\rangle_0$" for any bilinear form and then derived a contradiction from an inequality that M-invariance does not forbid; the argument above replaces it.)

The next result generalizes Macdonald's formula. An analogous result is [Wa89a, Lemma I.3.1]. Here we use Macdonald's formula for G and for M to evaluate the constants.

Lemma 3.3. *Let χ be unramified regular, $f' \in V_B(\chi^{-1})$ of level n, and $\epsilon = \epsilon(n)$. There are M-maps $\bar{R}_{M,w} : V_B(\chi^{-1}) \to i_{M,A}((w\chi)^{-1})$, $w \in [W_M\backslash W]$, factoring through $r_{M,\bar{P}}$, such that for $a \in A_P^-(\epsilon)$*

$$\langle \pi_\chi(a)\Phi_{K,\chi}, f'\rangle_G = \delta_P^{1/2}(a) \sum_{w\in[W_M\backslash W]} c_w(\chi)\,\langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi},\ \bar{R}_{M,w}f'\rangle_M,$$

where $\Phi^M_{w\chi}$ is the normalized $(K\cap M)$-fixed vector of $i_{M,A}(w\chi)$. For $f' = \Phi_{K,\chi^{-1}}$ one has $\bar{R}_{M,w}\Phi_{K,\chi^{-1}} = c(w, M, \chi)\, c_w(\chi^{-1})\, \Phi^M_{(w\chi)^{-1}}$ with

$$c(w, M, \chi) = Q_G^{-1}Q_M\, \frac{\prod_{\alpha\in\Sigma^+\smallsetminus\Sigma_M^+} c_\alpha\big((w\chi)^{-1}\big)}{c_w(\chi)\, c_w(\chi^{-1})},$$

and then the identity holds for all $a \in A^-$. In particular, for $M = M_0$, $c(w, M_0, \chi) = Q_G^{-1}c((w\chi)^{-1})/\big(c_w(\chi)c_w(\chi^{-1})\big)$, as in [Ma71].

Remark 4. Since K_n is normal in K, $\pi(k)f'$ has level n whenever f' does, so one $\epsilon = \epsilon(n)$ serves for all $\pi(k)f'$, $k \in K$; and since the vectors $u(w, k)$ of Lemma 3.4 have level $\le n$ in M, the same ϵ serves for the analogous identity on Levi subgroups.

Proof of Lemma 3.3. The first identity is (3.1) combined with Lemmas 3.1 and 3.2, with $(\Phi_{K,\chi})_w = R_M T_w \Phi_{K,\chi} = c_w(\chi)\Phi^M_{w\chi}$ by (2.8); the scalars of Lemma 3.2 are absorbed into $\bar{R}_{M,w}$.

For the constants take $f' = \Phi_{K,\chi^{-1}}$. Then the left side is $\Gamma_\chi(a)$, and by $K\cap M$-invariance $\bar{R}_{M,w}\Phi_{K,\chi^{-1}}$ is a multiple of $\Phi^M_{(w\chi)^{-1}}$; write the multiple as $c(w, M, \chi)c_w(\chi^{-1})$. Since

$\langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi}, \Phi^M_{(w\chi)^{-1}}\rangle_M = \Gamma^M_{w\chi}(a)$, Macdonald's formula for M (Lemma 2.1 for M, valid on $A^- \subset A^-_M$) gives, with $\delta_P^{1/2}\delta_{B\cap M}^{1/2} = \delta_B^{1/2}$,

$$\mathrm{RHS} = Q_M^{-1}\delta_B^{1/2}(a) \sum_{w\in[W_M\backslash W]} c(w,M,\chi)c_w(\chi)c_w(\chi^{-1}) \sum_{v\in W_M} c_M\big((vw\chi)^{-1}\big)(vw\chi)(a),$$

while Macdonald's formula for G gives $\Gamma_\chi(a) = Q_G^{-1}\delta_B^{1/2}(a)\sum_{\sigma\in W} c((\sigma\chi)^{-1})(\sigma\chi)(a)$. Every $\sigma \in W$ is uniquely $\sigma = vw$ with $v \in W_M$, $w \in [W_M\backslash W]$, and the characters $a \mapsto (\sigma\chi)(a)$ are linearly independent on $A_P^-(\epsilon)$ for regular χ. Comparing coefficients,

$$c(w,M,\chi) = Q_G^{-1}Q_M \frac{c((vw\chi)^{-1})}{c_M((vw\chi)^{-1})\, c_w(\chi)c_w(\chi^{-1})} \qquad (v \in W_M).$$

The ratio $c(\eta)/c_M(\eta) = \prod_{\alpha\in\Sigma^+\smallsetminus\Sigma_M^+} c_\alpha(\eta)$ is unchanged when η is replaced by $v\eta$, because W_M permutes $\Sigma^+ \smallsetminus \Sigma_M^+$; this gives the formula. Since both Macdonald formulas hold on all of A^-, so does the identity for spherical vectors. □

Remark 2026.5 (10 Nov. 2026) (Lemma 3.3). The 1997 statement paired R_MT_wf with R_MT_wf' for *arbitrary* f, f', omitted $\delta_P^{1/2}(a)$, was stated on $A^-(\epsilon)$, and had $c(w\chi)$ where (with the corrected Lemma 2.1) $c((w\chi)^{-1})$ belongs. For general f' the functional $f' \mapsto \langle\pi(a)\Phi, f'\rangle$ factors through the $\bar{P}$-Jacquet module, not the P-Jacquet module, so we have stated the lemma with $\bar{R}_{M,w}$. Identifying $\bar{R}_{M,w}$ with "R_MT_w" requires Waldspurger's opposite-parabolic conventions, which is presumably the origin of the factor $w_\ell^G w_\ell^M$ in the 1997 text (log item 3.61). Only the spherical case is used explicitly below.

Lemma 3.4. *Let χ, χ' be regular unramified characters, $u \in V_B(\chi^{-1})$ and $u' \in V_B(\chi'^{-1})$ of level n, and $a \in A_P^-(\epsilon(n))$. Then*

$$\int_{KaK} \langle\pi_\chi(g)\Phi_{K,\chi}, u\rangle_G\langle\pi_{\chi'}(g)\Phi_{K,\chi'}, u'\rangle_G\, dg = \mathrm{meas}(KaK)\,\delta_P(a) \sum_{w,w'\in[W_M\backslash W]} c_w(\chi)c_{w'}(\chi')$$
$$\times \int_K \langle i_{M,A}(w\chi)(a)\Phi^M_{w\chi}, u(w,k)\rangle_M\langle i_{M,A}(w'\chi')(a)\Phi^M_{w'\chi'}, u'(w',k)\rangle_M\, dk,$$

where $u(w,k) = \bar{R}_{M,w}(\pi(k)u)$ and $u'(w',k) = \bar{R}_{M,w'}(\pi(k)u')$ (with $\bar{R}$ formed for χ, resp. χ').

Remark 5. If κ is an unramified character of $G(F)$ then κ is constant on KaK, so the integral above with an extra factor $\kappa(g)$ equals $\kappa(a)$ times the integral without it. This is used for Proposition 3.7.

Proof of Lemma 3.4. The push-forward of Haar measure on $K \times K$ under $(k_1, k_2) \mapsto k_1ak_2$ is $\mathrm{meas}(KaK)^{-1}$ times Haar measure on KaK. Since $\Phi_{K,\chi}$ is K-fixed, $\langle\pi_\chi(k_1ak_2)\Phi_{K,\chi}, u\rangle_G = \langle\pi_\chi(a)\Phi_{K,\chi}, \pi(k_1^{-1})u\rangle_G$. Substituting $k = k_1^{-1}$, the left side is

$$\mathrm{meas}(KaK)\int_K \langle\pi_\chi(a)\Phi_{K,\chi}, \pi(k)u\rangle\langle\pi_{\chi'}(a)\Phi_{K,\chi'}, \pi(k)u'\rangle\, dk$$

Now apply Lemma 3.3 to each factor, which is legitimate by Remark 4, and multiply out. □

3.2 Truncated inner products of matrix coefficients

Let $T \in X_*(A)$ with $d(T) := \min_{\alpha\in\Delta}\langle T,\alpha\rangle > 0$. The set

$$\mathcal{A}_A(T) = \big\{\lambda \in X_*(A)/X_*(A_G) :\ \langle\lambda,\alpha\rangle \geq 0,\ \langle\lambda - T, \varpi_\alpha\rangle < 0\ \ \forall\alpha\in\Delta\big\} \tag{3.3}$$

is finite. Using (2.5), define

$$J^T(\chi,\chi',u,u') = \sum_{\lambda\in\mathcal{A}_A(T)} \int_{Ka_\lambda K} \langle\pi_\chi(g)\Phi_{K,\chi}, u\rangle_G \langle\pi_{\chi'}(g)\Phi_{K,\chi'}, u'\rangle_G\, dg, \tag{3.4}$$

and put $G(T) = \bigcup_{\lambda\in\mathcal{A}_A(T)} Ka_\lambda K$ as in (4.2).

Remark 2026.6 (10 Nov. 2026) (a false claim deleted). The 1997 text asserted that T can be chosen so that $\nu_A^{-1}(\mathcal{A}_A(T)^c) \subset A^-(\epsilon)$, i.e. that everything outside the truncated region is deep in every simple direction. This is false: in $SL(3)$, $\lambda_n = n(2\alpha_1^\vee + \alpha_2^\vee)$ lies outside $\mathcal{A}_A(T)$ for large n while $\langle\lambda_n,\alpha_2\rangle = 0$. The proof of Proposition 3.6 below avoids the claim by decomposing the dominant cone into regions on each of which some $A_P^-(\epsilon)$-asymptotic applies.

The vertex decomposition. For a standard parabolic Q let T_Q denote the projection of T to $\mathfrak{a}_Q$. Let

$$L_0 = \{T \in X_*(A) :\ T_Q \in X_*(A) \text{ for every standard } Q\},$$

a sublattice of finite index in $X_*(A)$. If G is semisimple, $T_G = 0$ and $T_B = T$.

Lemma 3.5A (vertex decomposition; replaces Lemma 3.5)**.** *Assume G semisimple (in general, work modulo $X_*(A_G)$). Let ψ be a character of $X_*(A)$, i.e. a point of $\widehat{T}$, and let $\mathsf{c} : X_*(A)^+ \to \mathbb{C}$ be a function that depends only on the set $\{\alpha\in\Delta : \langle\lambda,\alpha\rangle = 0\}$. For each standard parabolic Q there is a rational function $R_Q = R_Q^{\mathsf{c}}$ on $\widehat{T}$, independent of T, such that for all $T \in L_0$ with $d(T) > 0$*

$$\sum_{\lambda\in\mathcal{A}_A(T)} \mathsf{c}(\lambda)\,\psi(\lambda) = \sum_{Q\supseteq B} \psi(T_Q)\,R_Q(\psi) \tag{3.5A}$$

as rational functions of ψ. The denominator of R_Q is a product of factors $1 - \psi(\mu)$, where $\mu \in X_*(A)\smallsetminus\{0\}$ runs over primitive vectors on the edges of the cone

$$C_Q = \{X \in \mathfrak{a}_0 :\ \langle X,\alpha\rangle \geq 0\ (\alpha\in\Delta^Q),\ \ \langle X,\varpi_\alpha\rangle \leq 0\ (\alpha\in\Delta\smallsetminus\Delta^Q)\},$$

whose edges are spanned by the fundamental coweights of M_Q (in $\mathfrak{a}_0^Q$) and by the vectors $-(\alpha^\vee)_Q$, $\alpha\in\Delta\smallsetminus\Delta^Q$. The term for $Q = G$ is independent of T: $R_G(\psi)$ is the meromorphic continuation of $\sum_{\lambda\in X_*(A)^+}\mathsf{c}(\lambda)\psi(\lambda)$.

Proof. Let $\Pi_T = \{X\in\mathfrak{a}_0 : \langle X,\alpha\rangle \geq 0,\ \langle X - T,\varpi_\alpha\rangle \leq 0\ \forall\alpha\}$; this is the dominant part of the convex hull of $W\cdot T$. *Vertices.* For Q standard, T_Q satisfies $\langle T_Q,\alpha\rangle = 0$ for $\alpha\in\Delta^Q$ (since $T_Q\in\mathfrak{a}_Q$) and $\langle T_Q - T,\varpi_\alpha\rangle = 0$ for $\alpha\notin\Delta^Q$ (since such ϖ_α vanish on $\mathfrak{a}_0^Q$). Writing $T^Q = \sum_{\beta\in\Delta^Q} t_\beta\beta^\vee$ with $t_\beta\geq 0$ (as T^Q is dominant for M_Q) and using $\langle\beta^\vee,\alpha\rangle\leq 0$ for $\alpha\neq\beta$, one finds $\langle T_Q,\alpha\rangle \geq \langle T,\alpha\rangle > 0$ for $\alpha\notin\Delta^Q$ and $\langle T_Q - T,\varpi_\alpha\rangle = -t_\alpha < 0$ for $\alpha\in\Delta^Q$. So T_Q is a vertex of Π_T, with active constraints exactly those defining C_Q, and Π_T is simple there. Conversely, if v is a vertex and $\theta = \{\alpha : \langle v,\alpha\rangle = 0\}$, then $v\in\mathfrak{a}_{P_\theta}$. For $\alpha\in\theta$ the restriction of ϖ_α to $\mathfrak{a}_{P_\theta}$ is a combination $\sum_{\beta\notin\theta} c_{\alpha\beta}\varpi_\beta$ with $c_{\alpha\beta} = \langle(\beta^\vee)_{P_\theta},\varpi_\alpha\rangle \geq 0$. Hence the constraints $\langle v - T,\varpi_\alpha\rangle\leq 0$ for $\alpha\in\theta$ are strict, those for $\beta\notin\theta$ must all be active, and $v = T_{P_\theta}$.

Counting. $\mathsf{c}\cdot\mathbf{1}_{\mathcal{A}_A(T)}$ is a finite linear combination, with coefficients independent of T, of indicator functions of closed faces of Π_T: write each relatively open face of the dominant cone, and each strict inequality $\langle\lambda - T, \varpi_\alpha\rangle < 0$, by inclusion–exclusion over closed faces. Each closed face Π' is a rational polytope whose vertices lie among the T_Q, with tangent cones that are faces of the C_Q and do not depend on T. By Brion's theorem [Br88] (for rational polytopes, see [Ba08]), $\sum_{\lambda\in\Pi'\cap X_*(A)}\psi(\lambda) = \sum_v \sigma_{v+C'_v}(\psi)$, the sum of the generating functions of the lattice points of the tangent cones at the vertices v of Π'. For $v = T_Q \in X_*(A)$ (here we use $T \in L_0$), $\sigma_{T_Q+C'}(\psi) = \psi(T_Q)\sigma_{C'}(\psi)$. Collecting terms gives (3.5A), with R_Q a T-independent combination of the $\sigma_{C'}$ for faces C' of C_Q. The denominators are those of simplicial-cone generating functions: $\sigma_{C'} = (\text{Laurent polynomial})/\prod(1-\psi(\mu_i))$ over primitive edge vectors μ_i. The edges of C_Q: an edge lies in $\mathfrak{a}_0^Q$ and is M_Q-dominant extremal (a fundamental coweight of M_Q), or lies in $\mathfrak{a}_Q$ and is dual to $\hat{\Delta}_Q$, i.e. is $-(\alpha^\vee)_Q$ for some $\alpha \notin \Delta^Q$. For $Q = G$ the only vertex is 0 and the cone is $X_*(A)^+$. □

Remark 2026.7 (10 Nov. 2026) (Lemma 3.5A). The 1997 Lemma 3.5 asserted that the ω-th term carries the factor $\prod_\alpha \chi(a_\alpha)^{T_\alpha}$ times a T-independent function; this is false already for $SL(2)$, where the regular part is $(z - z^T)/(1 - z)$. It also used coordinates $\langle a, \alpha\rangle$ on $X_*(A)$, which are unimodular only for adjoint G. Lemma 3.5A is the correct form. The vertex description and the edge directions were checked by hand; the inclusion–exclusion bookkeeping for half-open faces is standard but was written out in full only in rank one. In rank one ($X_*(A) = \mathbb{Z}\varpi^\vee$, $z = \psi(\varpi^\vee)$, $\mathsf{c}\equiv 1$): $\sum_{0\le n<T} z^n = \frac{1}{1-z} - \frac{z^T}{1-z}$, i.e. $R_G = \frac{1}{1-z}$, $R_B = -\frac{1}{1-z}$.

From [Ma71, Prop. 3.2.15] we have, for $a \in A^-$,

$$\mathrm{meas}(KaK) = \delta_B(a)^{-1}Q_a, \qquad Q_a = Q_G/Q_{W_a}, \tag{3.7}$$

where $W_a \subset W$ is the stabilizer of $\nu_A(a)$ (generated by the w_α with $\langle\nu_A(a), \alpha\rangle = 0$) and $Q_{W_a} = \sum_{w\in W_a}(IwI : I)^{-1}$. So Q_a is a rational number (for $SL(2)$ and $a \neq 1$, $Q_a = 1 + q^{-1}$), depending only on the face of the dominant cone containing $\nu_A(a)$. The 1997 text wrote $Q_a \in \mathbb{Z}$.

The case of Lemma 3.5A needed for K-fixed vectors is this. For K-fixed $u = \Phi_{K,\chi^{-1}}$, $u' = \Phi_{K,\chi'^{-1}}$, Lemma 3.3 holds on all of A^-, and Lemma 3.4 with $P = B$ gives

$$J^T = \sum_{w,w'} C_{w,w'} \sum_{\lambda\in\mathcal{A}_A(T)} Q_{a_\lambda}\big((w\chi)(w'\chi')\big)(\lambda),$$

where

$$C_{w,w'} = c_w(\chi)c_{w'}(\chi')c(w, M_0, \chi)c(w', M_0, \chi')c_w(\chi^{-1})c_{w'}(\chi'^{-1}).$$

By Lemma 3.5A with $\mathsf{c}(\lambda) = Q_{a_\lambda}$ and $\psi = (w\chi)(w'\chi')$, this equals

$$\sum_{w,w'\in W} C_{w,w'} \sum_{Q\supseteq B} \big((w\chi)(w'\chi')\big)(T_Q)\, R_Q\big((w\chi)(w'\chi')\big). \tag{3.8}$$

This replaces the 1997 formula (3.8), which used the incorrect Lemma 3.5.

The main proposition.

Proposition 3.6 (2026 form)**.** *Fix a level n. There are $c = c(n) \ge 1$ and $d_0 = d_0(n)$ with the following property. Let χ, χ' be unramified regular characters and $u \in V_B(\chi^{-1})$, $u' \in V_B(\chi'^{-1})$*

of level n. Then for $T \in L_0$ with $d(T) \geq d_0$,

$$J^T(\chi,\chi',u,u') = \sum_{Q \supseteq B} J_Q^T(\chi,\chi',u,u'), \qquad J_Q^T = \sum_{P \subseteq Q} \sum_{w,w' \in [W_{M_P} \backslash W]} \psi_{w,w'}(T_Q)\, \mathcal{R}_{w,w'}^{P,Q}(\chi,\chi';u,u'), \tag{3.9}$$

where $\psi_{w,w'} = (w\chi)(w'\chi')$, and each $\mathcal{R}_{w,w'}^{P,Q}$ is bilinear in (u,u'), rational in (χ,χ') and independent of T. In particular $J_G^T(\chi,\chi',u,u') =: J_G(\chi,\chi',u,u')$ does not depend on T; it is the "T-independent remainder", and it contains the contribution of the compact core where no asymptotics are available. For K-fixed u,u', (3.9) is (3.8).

Proof. Let $\epsilon = \epsilon(n)$ be as in (3.1) and choose an integer $c \geq 1$ with $q^{-c} \leq \epsilon$.

Step 1: regions. For a standard parabolic $P = MN$ let

$$S_P = \{\lambda \in X_*(A)^+ :\ \langle \lambda, \alpha \rangle < c \iff \alpha \in \Delta^M\}.$$

These sets partition $X_*(A)^+$, and $a_\lambda \in A_P^-(\epsilon)$ for $\lambda \in S_P$. The map $\lambda \mapsto \lambda^P$ identifies $X_*(A)/X_*(A_M)$ with a lattice in $\mathfrak{a}_0^P$, and for $\lambda \in S_P$, λ^P lies in the bounded set $\{X \in \mathfrak{a}_0^P : 0 \leq \langle X, \alpha \rangle < c,\ \alpha \in \Delta^M\}$. So S_P meets only finitely many cosets of $X_*(A_M)$. Choose representatives $e \in S_P$ for them, a finite set E_P. Then

$$S_P \cap (e + X_*(A_M)) = e + \{\nu \in X_*(A_M) :\ \langle e+\nu, \alpha \rangle \geq c \ \ (\alpha \in \Delta \smallsetminus \Delta^M)\}.$$

Step 2: exponential form on each region. Let $F(\lambda)$ denote the λ-th summand of (3.4). For $\lambda = e + \nu \in S_P$, apply Lemma 3.4. We have $a_\lambda = a_e a_\nu$ with $a_\nu \in A_M(F)$ central in M, so $i_{M,A}(w\chi)(a_\lambda) = (w\chi)(\nu)\, i_{M,A}(w\chi)(a_e)$. Further $\mathrm{meas}(Ka_\lambda K)\delta_P(a_\lambda) = \delta_{B\cap M}(a_\lambda)^{-1} Q_{a_\lambda} = \delta_{B\cap M}(a_e)^{-1} Q_{a_e}$, because $\delta_{B\cap M}$ is trivial on A_M and the stabilizer W_{a_λ} only involves roots in Δ^M, where $\langle \nu, \alpha \rangle = 0$. Hence

$$F(e+\nu) = \sum_{w,w' \in [W_M \backslash W]} \psi_{w,w'}(\nu)\, C_{w,w',e}^P(u,u'),$$

$$C_{w,w',e}^P(u,u') = \frac{Q_{a_e} c_w(\chi) c_{w'}(\chi')}{\delta_{B\cap M}(a_e)} \int_K \langle i_{M,A}(w\chi)(a_e)\Phi_{w\chi}^M, u(w,k)\rangle_M \\ \times \langle i_{M,A}(w'\chi')(a_e)\Phi_{w'\chi'}^M, u'(w',k)\rangle_M \, dk.$$

Step 3: the truncated region on $S_P \cap (e + X_*(A_M))$. The conditions $\langle e + \nu - T, \varpi_\alpha \rangle < 0$ for $\alpha \in \Delta^M$ are implied by those for $\alpha \notin \Delta^M$ once $d(T)$ is large. Indeed, for $Y \in \mathfrak{a}_P$ and $\alpha \in \Delta^M$, $\varpi_\alpha(Y) = \sum_{\beta \notin \Delta^M} c_{\alpha\beta} \varpi_\beta(Y)$ with $c_{\alpha\beta} = \varpi_\alpha((\beta^\vee)_P) \geq 0$ (proof of Lemma 3.5A). So if $\varpi_\beta(e+\nu) < \varpi_\beta(T)$ for all $\beta \notin \Delta^M$, then

$$\varpi_\alpha(e+\nu) = \varpi_\alpha(e^P) + \textstyle\sum_\beta c_{\alpha\beta}\varpi_\beta(e+\nu) < \varpi_\alpha(e^P) + \varpi_\alpha(T_P) = \varpi_\alpha(T) - \varpi_\alpha(T^P) + \varpi_\alpha(e^P),$$

and $\varpi_\alpha(T^P) \geq \varpi_\alpha(e^P)$ as soon as $d(T) \geq d_0$, since $\varpi_\alpha(T^P)$ grows linearly in $d(T)$ while E_P is finite. (For $P = G$, $\nu = 0$ and the condition is $e \in \mathcal{A}_A(T)$, true for all $e \in E_G$ once $d(T) \geq d_0$.) Thus the ν that occur form the lattice points of the half-open polytope

$$\Pi_{P,e}(T) = \{\nu \in \mathfrak{a}_P :\ \langle \nu, \alpha \rangle \geq c - \langle e, \alpha \rangle \ (\alpha \in \Delta \smallsetminus \Delta^M),\ \ \langle \nu - (T-e), \varpi_\beta \rangle < 0 \ (\beta \in \Delta \smallsetminus \Delta^M)\},$$

which has the shape of (3.3) for the root datum $(\mathfrak{a}_P, \Delta_P, \hat{\Delta}_P)$. Exactly as in the proof of Lemma 3.5A, its vertices are the points $v_{Q,e}(T)$, $Q \supseteq P$, determined by $\langle v, \alpha \rangle = c - \langle e, \alpha \rangle$

for $\alpha \in \Delta^Q \smallsetminus \Delta^M$ and $\langle v-(T-e),\varpi_\beta\rangle = 0$ for $\beta \notin \Delta^Q$. Decomposing $v = v_Q + v^Q$ in $\mathfrak{a}_P = \mathfrak{a}_Q \oplus \mathfrak{a}_P^Q$ gives $v_Q = T_Q - e_Q$, and $v^Q \in \mathfrak{a}_P^Q$ is determined by T-independent equations. Thus $v_{Q,e}(T) = T_Q + x_{Q,e}$ with $x_{Q,e}$ independent of T. Since $T_Q \in X_*(A)\cap\mathfrak{a}_Q \subset X_*(A_M)$ for $T \in L_0$, Brion's theorem gives

$$\sum_{\nu\in\Pi_{P,e}(T)\cap X_*(A_M)} \psi(\nu) = \sum_{Q\supseteq P} \psi(T_Q)\, R_{P,Q,e}(\psi)$$

with T-independent rational $R_{P,Q,e}$.

Step 4. Summing F over $\mathcal{A}_A(T) = \bigsqcup_P \bigsqcup_{e\in E_P}\big(\mathcal{A}_A(T)\cap(e+X_*(A_M))\big)$ gives (3.9) with $\mathcal{R}^{P,Q}_{w,w'} = \sum_{e\in E_P} C^P_{w,w',e}\, R_{P,Q,e}(\psi_{w,w'})$. For K-fixed vectors we may take $\epsilon = 1$, and the computation reduces to (3.8). □

Remark 2026.8 (10 Nov. 2026) (Proposition 3.6). The 1997 Proposition 3.6 claimed $J^T = \sum_{w,w'\in W} C^T(\chi,\chi',w,w')\langle T_w u, T_{w'}u'\rangle_G$ with coefficients independent of u,u', and that C^T could be computed from Lemma 3.5. Its inductive proof (equations (3.9)–(3.23) of 1997, via the functions $\xi(M,D,T,U)$ and Langlands' combinatorial lemma) shows only that $J^T - F(T) = J^U - F(U)$ for an explicit F. That leaves a T-independent constant undetermined (the compact-core contribution), and the structural claim for arbitrary u,u' is unsupported (log items 3.46, 3.69). The proof above is new (2026). It keeps the ingredients of the 1997 argument (Casselman's asymptotics on the regions $A_P^-(\epsilon)$, Macdonald's formula, and a combinatorial identity for truncated cones), but uses a direct partition of the dominant cone together with Brion's theorem instead of the induction on Levi subgroups. It does not determine the remainder J_G, which is genuinely non-zero (in the $PGL(2)$ example it produces $|B_r| = O_\gamma(\mathbf{1}_K)$, §6.1). Steps 1–4 have been checked in rank one against §6; in higher rank they have been checked on paper only.

Proposition 3.7. *Let κ be an unramified character of $G(F)$. Then $w\kappa = \kappa$ for all $w \in W$ (represented in K), and the analogue $J^{T,\kappa}$ of (3.4) with an extra factor $\kappa(g)$ in the integrand has the decomposition (3.9) with $\psi_{w,w'}$ replaced by $\kappa|_{M_0(F)}\cdot\psi_{w,w'}$.*

Proof. By Remark 5 the extra factor is $\kappa(a_\lambda)$ on the λ-th summand, and $\lambda\mapsto\kappa(a_\lambda)$ is a character of $X_*(A)$; so the proof of Proposition 3.6 applies verbatim. □

4 Integrating the kernel

4.1 $(G,A)^\sim$-families

We prove in this section an analogue of [Wa89a, Prop. II.5]. For $P\in\mathcal{P}(A)$ let $w_P\in W$ be the element with $P = w_P^{-1}Bw_P$. The 1997 text introduced, for $\phi\in C_c^\infty(G(F))$,

$$Q^T(\chi,\chi',w,w',\phi) = \operatorname{tr}\left[T_w^* T_{w'}\,\pi_\chi(\phi)\right]\, C^T(\chi,\chi'^{-1},w,w'), \tag{4.1}$$

where T_w^* is the transpose of T_w (an operator $V_B(w\chi)\to V_B(\chi)$ in the compact picture) and C^T is the coefficient of the 1997 Proposition 3.6. (The 1997 definition also contained a factor $\tilde\theta(w\chi/w'\chi')$, which Proposition 4.2 then divided out again.) It asserted:

Lemma 4.1 (1997)**.** *The collection $\{Q^T(\chi,\chi',w_P,w',\phi) : P\in\mathcal{P}(A)\}$ multiplied by $\tilde\theta(w_P\chi/w'\chi')$ forms a $(G,A)^\sim$-family in χ, and similarly in χ'.*

The sketched proof used that $\lambda \mapsto \mathrm{tr}[R_{wP}\, i_{G,A}(\chi_\lambda)(\phi)]$ (R_w the normalized intertwining operator) is a (G,A)-family [Ar81, §7], and that the product of two $(G,A)^\sim$-families is one.

Remark 2026.9 (10 Nov. 2026) (Lemma 4.1 and regularity). Since the 2026 Proposition 3.6 no longer produces coefficients of the form $C^T\langle T_w u, T_{w'}u'\rangle$, Lemma 4.1 is not used below, and it has not been verified. Its purpose in 1997 was to show that the singularities of the individual vertex terms cancel in the sum over P. This kind of cancellation does hold: in the $SL(2)$ example (§6.2) the sum of the vertex terms at the vertex T is a smooth function of χ for each T. But it is not enough for removing the truncation. The sum is smooth for each T but not uniformly in T: near the bad point it behaves like a Dirichlet kernel and tends to a point mass as $T \to \infty$. What is needed is regularity of each oscillatory component separately, which is Hypothesis (R) below. In particular, the argument proposed in the 2026 referee report (M8(b): smoothness of the $(G,A)^\sim$-sum plus the Riemann–Lebesgue lemma) does not suffice.

4.2 The Fourier transform of a truncated orbital integral

Let

$$G(T) := \bigcup_{\lambda \in \mathcal{A}_A(T)} K a_\lambda K, \tag{4.2}$$

and recall

$$\Gamma_\chi(g^{-1}hg) = \langle \pi_\chi(h)\pi_\chi(g)\Phi_{K,\chi},\ \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}\rangle_G. \tag{4.3}$$

For $\phi \in C_c^\infty(G(F))$ we define the *truncated orbital integral* (the Fourier transform of the truncated orbital integral of ϕ)

$$I^T(\chi,\phi) = \int_{G(F)^1} \Gamma_\chi(h) \int_{G(T)} \phi(g^{-1}hg)\, dg\, dh = \int_{G(T)\times G(F)^1} \Gamma_\chi(g^{-1}hg)\phi(h)\, dh\, dg. \tag{4.4}$$

In §5 we also consider

$$I^{T,\kappa}(\chi,\phi) := \int_{G'(T)\times G(F)} \kappa(g)\Gamma_\chi(g^{-1}hg)\phi(h)\, dh\, dg, \tag{4.5}$$

where $G = SL(n)$, $G' = PGL(n)$, κ is an unramified character of $G'(F)$, and $G'(T)$ is the set (4.2) formed for G' (with $K' = PGL(n,\mathcal{O})$ and the same T, viewed in $X_*(A')\otimes\mathbb{Q}$).

The idea is to expand (4.3) using a basis and, term by term, to use the computations of §3.

Proposition 4.2 (2026 form). *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$ be bi-K_n-invariant and let χ be unramified and regular. Let $\{u_i\}$ be a basis of $V_B(\chi)^{K_n}$ and $\{u_i^*\}$ the dual basis of $V_B(\chi^{-1})^{K_n}$.*

(a) For every T, with finite sums,

$$I^T(\chi,\phi) = \sum_{i,j} J^T(\chi,\chi^{-1},u_i^*,u_j)\,\langle \pi_\chi(\phi)u_i, u_j^*\rangle_G. \tag{4.7}$$

(b) There is a finite set $\mathcal{M} = \mathcal{M}(G,n)$ of linear maps $\mu : L_0 \to X_*(A)$, containing 0 and independent of χ and ϕ, such that for $T \in L_0$ with $d(T) \ge d_0(n)$

$$I^T(\chi,\phi) = \sum_{\mu\in\mathcal{M}} \chi(\mu(T))\, P_\mu(T;\chi,\phi), \tag{4.7A}$$

where $P_\mu(\cdot\,;\chi,\phi)$ is a polynomial function on L_0 whose coefficients are rational functions of χ, regular at generic regular χ, and linear in ϕ. The maps μ have the form $T \mapsto w^{-1}T_Q - w'^{-1}T_Q$ with Q standard and $w,w' \in W$.

(c) Define

$$I(\chi,\phi) := P_0(0;\chi,\phi),$$

the constant term of the non-oscillating part of $I^T(\chi,\phi)$. For generic regular χ this is intrinsic: it is the constant coefficient of the trivial character in the unique expansion of $T \mapsto I^T(\chi,\phi)$, on $\{T \in L_0 : d(T) \geq d_0\}$, as an exponential polynomial. The map $\phi \mapsto I(\chi,\phi)$ is linear and invariant under conjugation by K.

Proof. (a) Since ϕ is bi-K_n-invariant, $\pi_\chi(\phi)$ kills the complement of $V_B(\chi)^{K_n}$ and takes values in $V_B(\chi)^{K_n}$. So $\pi_\chi(\phi)v = \sum_{i,j}\langle v, u_i^*\rangle\langle \pi_\chi(\phi)u_i, u_j^*\rangle u_j$ for every v. Apply this to $v = \pi_\chi(g)\Phi_{K,\chi}$ in (4.3), integrate over h against ϕ and over $g \in G(T)$, and use $\langle u_j, \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}\rangle = \langle \pi_{\chi^{-1}}(g)\Phi_{K,\chi^{-1}}, u_j\rangle$. Every vector involved has level n, independently of T. (The 1997 version expanded $\pi(g)\Phi_{K,\chi}$ itself, whose level grows with g, and had u_j^* in place of u_j in (4.7).)

(b) Apply Proposition 3.6 to each $J^T(\chi,\chi',u_i^*,u_j)$ with χ' near χ^{-1}. The frequencies $\psi_{w,w'}(T_Q) = (w\chi)(w'\chi')(T_Q)$ tend to $\chi(\mu(T))$ with $\mu(T) = w^{-1}T_Q - w'^{-1}T_Q$. Group the terms of (3.9) according to the limit map μ. The total is analytic in χ' at $\chi' = \chi^{-1}$ (it is a finite sum of matrix-coefficient integrals over the compact set $G(T)$). For generic regular χ the characters $T \mapsto \chi(\mu(T))$, $\mu \in \mathcal{M}$, of L_0 are pairwise distinct. So by the uniqueness of exponential-polynomial expansions on a translated cone of L_0, each group extends analytically to $\chi' = \chi^{-1}$, and its value there is $\chi(\mu(T))$ times a polynomial in T. These polynomials are the confluent limits of the coalescing exponentials. Rationality in χ is inherited from Proposition 3.6.

(c) Intrinsic: by uniqueness of the expansion. Linearity is clear from (4.7). For $k \in K$, conjugating ϕ by k amounts to translating $g \mapsto gk$ in (4.4), and $G(T)$ is right K-invariant. □

Remark 2026.10 (10 Nov. 2026) (Proposition 4.2). The 1997 Proposition 4.2 stated $I(\chi,\phi) = \lim_{\chi'\to\chi}\sum_{w,w'} Q^T(\chi,\chi',w,w',\phi)\tilde{\theta}(w\chi/w'\chi')^{-1}$ "independent of T". But $I^T(\chi,\phi)$ genuinely depends on T; the 1997 justification ("follows from Proposition 3.6") does not apply, and $I(\chi,\phi)$ was never defined independently of the displayed limit. Here $I(\chi,\phi)$ is defined as the T-independent part. Part (b) is a straightforward consequence of Proposition 3.6, except for the confluence step, which we have written out only in rank one. The 1997 claim that $\phi \mapsto I(\chi,\phi)$ is G-admissible in the sense of [HC78, §14] is plausible, since by (4.7) it is a finite combination of $\phi \mapsto \mathrm{tr}(A\,\pi_\chi(\phi))$ with finite-rank A, but it needs a bound uniform in n that we have not checked. Full G-invariance follows from Theorem 4.5(b) when Hypothesis (R) holds (Corollary 4.5A).

Proposition 4.3. *Let $G = SL(n)$, $G' = PGL(n)$, A' the diagonal torus of G', and κ an unramified character of $G'(F)$. For χ an unramified regular character of $M_0(F)$, the truncated κ-integral $I^{T,\kappa}(\chi,\phi)$ has an expansion (4.7A), obtained by applying Proposition 3.7 for G' to the restriction of Γ_χ to $G'(T)$-conjugates. We put $I^\kappa(\chi,\phi) := P_0^\kappa(0;\chi,\phi)$.*

Remark 9. We only prove Proposition 4.2; the proof of 4.3 is similar, using Remark 5 and Proposition 3.7, with $\kappa|_{A'(F)}\cdot\chi$ in place of χ in the frequencies.

4.3 The spectral expansion

Let ϕ be a locally constant function with compact support in G_{ell}. By [Ka86, §4, Lemma 6] the distribution

$$f \longmapsto I(f,\phi) = \int_{G(F)^1\times G(F)^1} f(g^{-1}hg)\phi(h)\,dg\,dh = \int_{G(F)^1} f(h)\Phi_\phi(h)\,dh = \int_{G(F)^1}\Phi_f(h)\phi(h)\,dh, \tag{4.8}$$

$f \in C_c^\infty(G(F))$, has compactly generated support. Here, for $h \in G_{\mathrm{ell}}$, $\Phi_\phi(h) := \int_{G(F)^1} \phi(g^{-1}hg)\,dg$, and Φ_f is defined in the same way. (In (4.8) f may be any function in $C_c^\infty(G(F))$; from Lemma 4.4 on, $f \in \mathcal{H}(G,K)$.)

Lemma 4.4. *Let ϕ be locally constant with compact support in G_{ell}, and let $f \in \mathcal{H}(G,K)$. There is a compact set $C_{f,\phi} \subset G(F)^1$ such that $f(g^{-1}hg)\phi(h) \neq 0$ implies $g \in C_{f,\phi}$. If $G = SL(n)$ and $G' = PGL(n)$, there is likewise a compact $C'_{f,\phi} \subset G'(F)$ for the action of $G'(F)$ by conjugation.*

Proof. For each $h \in G_{\mathrm{ell}}$ there are a neighborhood $V_{h,f}$ of h and a compact set $C_{h,f} \subset G(F)^1$ such that $f(g^{-1}h'g) \neq 0$ and $h' \in V_{h,f}$ imply $g \in C_{h,f}$. This uses that the centralizer of an elliptic element is compact modulo A_G. Finitely many $V_{h,f}$ cover $\mathrm{supp}(\phi)$. The second part is similar. □

Choose $T = T(f,\phi) \in L_0$ with $d(T) \geq d_0(n)$ and $G(T) \supseteq C_{f,\phi}$, so that

$$\mathrm{supp}_g\Big(\int_{G(F)^1} f(g^{-1}hg)\phi(h)\,dh \Big) \subset G(T). \tag{4.9}$$

Then $I(f,\phi) = \int_{G(T)\times G(F)^1} f(g^{-1}hg)\phi(h)\,dg\,dh$. For $f \in \mathcal{H}(G,K)$, Corollary 2.3 gives $f(g^{-1}hg) = \int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\,d\mu(\chi)$. Write $d\mu(\chi) = \rho(\chi)\,d\chi$ with $\rho = \frac{Q_G}{|W|}|c(\chi)|^{-2}$.

Hypothesis (R) (regularity, for G and level n)**.** For every bi-K_n-invariant $\phi \in C_c^\infty(G_{\mathrm{ell}})$ and every $\mu \in \mathcal{M}(G,n)$, each coefficient of the polynomial $P_\mu(T;\chi,\phi)$ of (4.7A), multiplied by $\rho(\chi)$, extends to a C^∞ function of $\chi \in \widehat{T}^u$.

Theorem 4.5 (2026 form)**.** *Let $f \in \mathcal{H}(G,K)$, let $\phi \in C_c^\infty(G_{\mathrm{ell}})$ be bi-K_n-invariant, and let $T(f,\phi)$ be as in (4.9).*

(a) For all $T \in L_0$ with $G(T) \supseteq G(T(f,\phi))$,

$$\begin{aligned} I(f,\phi) &= \sum_{\lambda \in \mathcal{A}_A(T)} \int_{Ka_\lambda K \times G(F)^1} \int_{\widehat{T}^u} f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\,d\mu(\chi)\,\phi(h)\,dg\,dh \\ &= \int_{\widehat{T}^u} f^\vee(\chi)\,I^T(\chi,\phi)\,d\mu(\chi). \end{aligned}$$

(b) If Hypothesis (R) holds for G and n, then

$$I(f,\phi) = \int_{\widehat{T}^u} f^\vee(\chi)\,I(\chi,\phi)\,d\mu(\chi),$$

the integral converging absolutely, with $I(\chi,\phi)$ as in Proposition 4.2(c).

Proof. (a) Fubini on the compact set $G(T) \times \mathrm{supp}\,\phi$, the integrand $f^\vee(\chi)\Gamma_\chi(g^{-1}hg)\phi(h)$ being bounded on $\widehat{T}^u \times G(T) \times \mathrm{supp}\,\phi$.

(b) By (a) and (4.7A), for $T \in L_0$ in the cone $\mathcal{C} = \{T : G(T) \supseteq G(T(f,\phi)),\ d(T) \geq d_0\}$,

$$I(f,\phi) = \sum_{\mu \in \mathcal{M}} \int_{\widehat{T}^u} f^\vee(\chi)\rho(\chi)\chi(\mu(T))P_\mu(T;\chi,\phi)\,d\chi.$$

Fix $T_2 \in \mathcal{C}$ and a regular dominant $T_1 \in L_0$ with $\mu(T_1) \neq 0$ for every $\mu \neq 0$ in $\mathcal{M}$ (possible, since $\mathcal{M}$ is finite). Put $T = T_2 + tT_1$, $t \in \mathbb{Z}_{\geq 0}$. For $\mu \neq 0$ the μ-term is a finite sum of terms

$p(t)\,\hat{g}(-\mu(T_2)-t\mu(T_1))$, where p is a polynomial and $\hat{g}$ denotes Fourier coefficients of a function $g = f^\vee \rho\, a$ which is C^∞ by (R). Fourier coefficients of smooth functions on a compact torus decay faster than any power, so these terms tend to 0 as $t \to \infty$. Hence the $\mu = 0$ term, which is a polynomial in t, converges to $I(f,\phi)$ as $t \to \infty$, so it is constant in t. Since T_1 ranges over an open cone of directions, the polynomial $T \mapsto \int f^\vee \rho\, P_0(T;\chi,\phi)\, d\chi$ on L_0 is constant along a spanning set of directions from every point of $\mathcal{C}$, hence constant, and equal to its value at $T = 0$. That value is $\int f^\vee(\chi) I(\chi,\phi)\, d\mu(\chi)$. Absolute convergence holds because $I(\cdot,\phi)\rho$ is smooth by (R). □

Corollary 4.5A. *Assume (R) for G and all n. Then $\phi \mapsto I(\chi,\phi)\rho(\chi)$ is, for almost every χ, a G-invariant distribution on G_{ell}, and for each $\gamma \in G_{\mathrm{ell}}$ there is a function $I(\cdot,\gamma)$ with $I(\cdot,\gamma)\rho$ smooth, such that (1.5) holds in the form*

$$\int_{G(F)^1} f(g^{-1}\gamma g)\, dg = \int_{\widehat{T}^u} f^\vee(\chi) I(\chi,\gamma)\, d\mu(\chi) \qquad (f \in \mathcal{H}(G,K)).$$

Proof. $I(f,\phi)$ is invariant under conjugation of ϕ, and the W-invariant Laurent polynomials $f^\vee$ are dense in $C(\widehat{T}^u/W)$; since $I(\cdot,\phi)\rho$ is smooth and W-invariant, it is determined by the numbers $I(f,\phi)$. Taking ϕ to be normalized characteristic functions of small neighborhoods of γ, on which orbital integrals of f are constant, gives the second statement. □

Remark 2026.11 (10 Nov. 2026) (Theorem 4.5).

1. Part (a) is the first line of the 1997 Theorem 4.5 and is unconditional. The 1997 second line, with $I(\chi,\phi)$ "independent of T", rested on the unsupported claim discussed in Remark 2026.10.
2. Part (b) is conditional on (R). Hypothesis (R) holds for $PGL(2)$ (§6.1). It fails for $SL(2)$ (§6.2), and there the conclusion of (b) is false: the spectral side needs an extra point mass. We expect (R) to hold whenever $Z(G)$ is connected (Conjecture 4.6(b)); for $GL(N)$ the analogous statement is the content of [Wa89a], which we have not re-checked against the formulation here.
3. In rank one, the proof of (b) can be sharpened: the oscillatory terms pair to zero *exactly* with $f^\vee$ once T is large, by a Fourier-support argument (§6.1).

When (R) fails. The singular behavior comes from the denominators $1 - \psi(\mu_e)$ of Lemma 3.5A, with $\psi = w\chi/w'\chi$, at points where ρ does not vanish to compensate. On such a hypersurface a term $\chi(\mu(T))a(\chi)$ with $a\rho$ having a simple pole behaves, as $T \to \infty$, like a Dirichlet kernel, and its pairing with $f^\vee$ tends to a multiple of the restriction of $f^\vee$ to the hypersurface (the Sokhotski–Plemelj formula on the circle). The limit exists because $I(f,\phi)$ does not depend on T. This motivates:

Conjecture 4.6. *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$.*

(a) The linear form $f^\vee \mapsto I(f,\phi)$ on Satake transforms is given by a measure dI_ϕ on $\widehat{T}^u/W$ of the form

$$dI_\phi = I(\chi,\phi)\, d\mu(\chi) + dI_\phi^{sing},$$

where $I(\cdot,\phi)$ is the function of Proposition 4.2(c), $I(\cdot,\phi)\rho \in L^1$, and dI_ϕ^{sing} is supported on the set of non-regular unitary χ at which ρ does not vanish (to the order forced by the stabilizer of χ). This set is, conjecturally, the set of χ for which the R-group of $i_{G,A}(\chi)$ is non-trivial.

(b) If $Z(G)$ is connected (equivalently, the derived group of $\hat{G}$ is simply connected), then the R-groups of the unitary unramified principal series are trivial [Ke82], Hypothesis (R) holds, and $dI_\phi^{sing} = 0$.

(c) For $G = SL(n)$, dI_ϕ^{sing} is the sum, over the non-trivial unramified κ, of the contributions $(c_S|\mathfrak{D}(S)|)^{-1}\, dI_\phi^\kappa$ of (5.4A), which are push-forwards of spectral measures of the elliptic endoscopic groups (Proposition 5.2).

Remark 2026.12 (10 Nov. 2026) (Conjecture 4.6). Verified for $PGL(2)$ (no singular part) and for $SL(2)$, where $dI_\gamma^{sing} = \frac{1}{2}\Delta\,\delta_{\chi_0}$ and χ_0 is the unramified quadratic character (§6.2). The precise support condition in (a) is a guess, and in (c) the normalizations (the size of $\mathfrak{D}$, transfer factors) have only been checked for $SL(2)$. Part (b) relies on [Ke82] (Chevalley groups; the general unramified case via Steinberg's connectedness theorem for centralizers in $\hat{G}$) for the triviality of R-groups; the implication "trivial R-groups $\Rightarrow$ (R)" is the conjectural part.

The geometric side. The Weyl integration formula states that

$$\int_{G(F)^1} \psi(h)\,dh = \sum_S \frac{1}{|W(G,S)|} \int_{S(F)\cap G(F)^1} |D(s)| \int_{S(F)\backslash G(F)^1} \psi(g^{-1}sg)\,d\dot{g}\,ds, \tag{4.10}$$

where S runs over a set of representatives of the $G(F)$-conjugacy classes of maximal F-tori, $W(G,S) = N_{G(F)}(S)/S(F)$, $D(s)$ is the Weyl discriminant, and $d\dot{g}$ is the quotient measure. (The 1997 text wrote "$\Delta(t)^2\,\mathrm{meas}(T(F))$" in one line and not in the next, and used T for these tori.) If ϕ is supported on G_{ell}, only elliptic S contribute. Plugging $\psi = f\Phi_\phi$ into (4.10) gives the "geometric" expansion

$$I(f,\phi) = \sum_{S\ \mathrm{ell}} \frac{\mathrm{vol}(S(F)/A_G(F))}{|W(G,S)|} \int_{S(F)\cap G(F)^1} |D(s)|\, O_s(f)\, O_s(\phi)\,ds, \tag{4.11}$$

where $O_s(f) = \int_{S(F)\backslash G(F)^1} f(g^{-1}sg)\,d\dot{g}$. The equality between Theorem 4.5 and (4.11) is a special case of Arthur's local trace formula (§7).

5 The κ-stable trace formula on $SL(n)$

5.1 A twisted trace formula on $SL(n,F)$

Let $G := SL(n)$, $G' := PGL(n)$, $m \mid n$, and let $\eta : F^\times \to \mathbb{C}^\times$ be an unramified character of order m. Then $\kappa'(g) := \eta(\det g)$ is a character of $GL(n,F)$, trivial on the center since $\eta(z^n) = 1$; let κ be the corresponding character of $G'(F)$. It is trivial on the image of $G(F)$, so for $f \in C_c^\infty(G(F))$ the *κ-orbital integral*

$$\Phi_f^\kappa(h) := \int_{G'(F)} \kappa(g) f(g^{-1}hg)\,dg \qquad (h \in G_{\mathrm{ell}})$$

is invariant under conjugation by $G(F)$, and $\Phi_f^\kappa(x^{-1}hx) = \kappa(x)^{-1}\Phi_f^\kappa(h)$ for $x \in G'(F)$. (The 1997 computation (5.3) wrote f for ϕ in the integrand, and its sign conventions for κ were inconsistent.)

Let f and ϕ be as in Lemma 4.4. Then

$$I^\kappa(f,\phi) := \int_{G(F)} \int_{G'(F)} \kappa(g) f(g^{-1}hg)\phi(h)\, dg\, dh = \int_{G(F)} \Phi_f^\kappa(h)\phi(h)\, dh = \int_{G(F)} \Phi_\phi^{\kappa^{-1}}(h) f(h)\, dh. \tag{5.1}$$

The last equality follows from the substitution $h \mapsto ghg^{-1}$ followed by $g \mapsto g^{-1}$, which turns κ into κ^{-1}. The interchanges of integration are justified by Lemma 4.4. For h regular elliptic, the properties of Φ_f^κ near the singular set of the centralizer were investigated in [Ka84] (see also [He86]).

Let $S \subset G$ be an elliptic maximal torus and S' its image in G'. The $G'(F)$-conjugacy class of a regular $s \in S(F)$ is the union of the $G(F)$-classes of the $x^{-1}sx$, $x \in \mathfrak{D}(S) := G'(F)/G(F)S'(F)$, a finite abelian group. Hence, with $c_S = \mathrm{vol}(S'(F))/\mathrm{vol}(S(F))$,

$$\Phi_f^\kappa(s) = c_S \sum_{x \in \mathfrak{D}(S)} \kappa(x)\, O_{x^{-1}sx}(f) \quad \text{if } \kappa|_{S'(F)} = 1, \qquad \Phi_f^\kappa(s) = 0 \quad \text{otherwise.} \tag{5.2}$$

Since Φ_f^κ is $G(F)$-invariant, the Weyl integration formula (4.10) for $G(F)$ gives

$$I^\kappa(f,\phi) = \sum_{S \text{ ell}} \frac{1}{|W(G,S)|} \int_{S(F)} |D(s)|\, \Phi_f^\kappa(s)\, O_s(\phi)\, ds. \tag{5.4}$$

This replaces (5.2)–(5.4) of 1997, which integrated over $S(F)\backslash G'(F)$ although the integrand lives on $G(F)$; the finite group $\mathfrak{D}(S)$ in (5.2) is the source of the κ-weights. Fourier inversion on $\mathfrak{D}(S)$ recovers ordinary orbital integrals:

$$O_s(f) = \frac{1}{c_S|\mathfrak{D}(S)|} \sum_{\kappa \in \mathfrak{D}(S)^\wedge} \Phi_f^\kappa(s). \tag{5.4A}$$

The method of proof of Theorem 4.5 applies to I^κ (Proposition 4.3). It gives an expression

$$I^\kappa(f,\phi) = \int_{\widehat{T}^u} f^\vee(\chi)\, dI_\phi^\kappa(\chi), \tag{5.7}$$

where A is the maximal split torus of G. Under (R) for $I^{T,\kappa}$, $dI_\phi^\kappa = I^\kappa(\chi,\phi)\, d\mu(\chi)$. Combining (5.7) and (5.4),

$$\int_{\widehat{T}^u} f^\vee(\chi)\, dI_\phi^\kappa(\chi) = \sum_{S \text{ ell}} \frac{1}{|W(G,S)|} \int_{S(F)} |D(s)|\, \Phi_f^\kappa(s)\, O_s(\phi)\, ds, \tag{5.8}$$

a special case of a κ-stable form of Arthur's local trace formula. For $\gamma \in G_{\mathrm{ell}}$ we write (5.7) as

$$\Phi_f^\kappa(\gamma) = \int_{\widehat{T}^u} f^\vee(\chi)\, dI_\gamma^\kappa(\chi), \qquad f \in \mathcal{H}(G,K). \tag{5.9}$$

5.2 The fundamental lemma and the spectral measures dI_γ^κ

For κ of order $m \mid n$ the corresponding elliptic endoscopic group of $SL(n)$ is

$$H = H_m = \ker\big(N_{E/F} \circ \det :\ \mathrm{Res}_{E/F} GL(n/m) \to \mathbb{G}_m\big), \qquad E/F \text{ unramified},\ [E:F] = m.$$

It is unramified over F and split over E. (The 1997 text wrote $\mathrm{Res}_{E/F}(GL(m)/E)$, and for $m=n$ "an elliptic Cartan of $GL(n)$". In fact, for $m=n$, H is the norm-one torus $\ker N_{E/F}$, an elliptic maximal torus of $SL(n)$.) Fix an L-embedding $\iota : {}^LH \hookrightarrow {}^LG$ and let $\iota^* : \mathcal{H}(G,K) \to \mathcal{H}(H,K_H)$ be the induced homomorphism, characterized by

$$(\iota^* f)^\vee(\chi_H) = f^\vee(\iota(\chi_H)),$$

where ι also denotes the induced map $\widehat{T}_H^u/W_H \to \widehat{T}^u/W$ on unramified unitary characters. The *fundamental lemma* for the spherical Hecke algebra asserts that for $\gamma \in G_{\mathrm{ell}}$ coming from a (G,H)-regular $\gamma_H \in H(F)$,

$$\Phi_f^\kappa(\gamma) = \Delta_{H,\kappa}(\gamma_H,\gamma)\, S\Phi^H_{\iota^* f}(\gamma_H), \tag{5.11}$$

where $S\Phi^H$ is the *stable* orbital integral on $H(F)$ and $\Delta_{H,\kappa}$ is the Langlands–Shelstad transfer factor [LS87]; and $\Phi_f^\kappa(\gamma)=0$ for γ not coming from H. This is now a theorem. For the unit element and $SL(n)$ it is due to Kazhdan [Ka84] (κ of order n) and Waldspurger [Wa91]. The extension from the unit element to the whole Hecke algebra is due to Hales [Ha95]. The general case is due to Ngô [Ng10], combined with Waldspurger's reduction from groups to Lie algebras. (The 1997 text said "it has been conjectured".)

Proposition 5.1. *Let κ have order n, so $H = \ker N_{E/F}$ is compact. Let $\chi_\kappa \in \widehat{T}^u/W$ be the image under ι of the trivial character of $H(F)$; its Satake parameter is the class of $\mathrm{diag}(1,\zeta,\dots,\zeta^{n-1})$ in $PGL(n,\mathbb{C})$, ζ a primitive n-th root of unity. For $\gamma \in G_{\mathrm{ell}}$ coming from $\gamma_H \in H(F)$,*

$$\Phi_f^\kappa(\gamma) = \Delta_{H,\kappa}(\gamma_H,\gamma)\, f^\vee(\chi_\kappa) \qquad (f \in \mathcal{H}(G,K)),$$

that is, $dI_\gamma^\kappa = \Delta_{H,\kappa}(\gamma_H,\gamma)\,\delta_{\chi_\kappa}$ is a point mass.

Proof. $H(F)$ is compact, so $\mathcal{H}(H,H(F))$ consists of constants, $S\Phi^H_{\iota^*f}(\gamma_H) = (\iota^*f)(\gamma_H)$ (with $\mathrm{vol}\, H(F)=1$), and $\iota^* f$ is the constant $(\iota^*f)^\vee(1) = f^\vee(\chi_\kappa)$. Now apply (5.11). Under ι, Frobenius is sent to an n-cycle in the Weyl group of $\hat{G} = PGL(n,\mathbb{C})$, whose semisimple class has eigenvalues the n-th roots of unity. □

Proposition 5.2. *Let $m \mid n$ and suppose that Theorem 4.5 holds for $H = H_m$ in its stable form: for $\gamma_H \in H_{\mathrm{ell}}$ there is a measure $dI^{H,st}_{\gamma_H}$ on $\widehat{T}_H^u/W_H$ with $S\Phi^H_{f_H}(\gamma_H) = \int f_H^\vee\, dI^{H,st}_{\gamma_H}$ for $f_H \in \mathcal{H}(H,K_H)$. Then for γ coming from γ_H,*

$$dI_\gamma^\kappa = \Delta_{H,\kappa}(\gamma_H,\gamma)\cdot \iota_*\big(dI^{H,st}_{\gamma_H}\big).$$

In words: the κ-spectral measure of G is the transfer factor times the push-forward, along the dual embedding, of the stable spectral measure of H.

Proof. Combine (5.11) with $(\iota^*f)^\vee = f^\vee\circ\iota$: $\Phi_f^\kappa(\gamma) = \Delta\int (f^\vee\circ\iota)\, dI^{H,st}_{\gamma_H} = \Delta\int f^\vee\, d\iota_*(dI^{H,st}_{\gamma_H})$. The measure dI_γ^κ is determined by its values on the dense set of Satake transforms. □

This answers the question with which the 1997 version closed: the distribution $\phi \mapsto I^\kappa(\iota(\chi_H),\phi)$ on G "transfers" to $\phi_H \mapsto I^{H,st}(\chi_H,\phi_H)$ in the sense of Proposition 5.2. By (5.4A), the ordinary spectral measure of $SL(n)$ is $dI_\gamma = (c_S|\mathfrak{D}(S)|)^{-1}\sum_\kappa dI_\gamma^\kappa$. The $\kappa=1$ term is the stable part, and the $\kappa\neq 1$ terms are push-forwards from endoscopic groups, supported on the proper subvarieties $\iota(\widehat{T}_H^u/W_H)$ of $\widehat{T}^u/W$. This is the source of the singular part in Conjecture 4.6(c). For $n=2$ it is the atom found in §6.2.

Remark 2026.13 (10 Nov. 2026) (§5). The structure of §5 (signs of κ, $\mathfrak{D}(S)$, the endoscopic groups H_m, the fundamental lemma as a theorem) is standard. The normalizations (c_S, the measures on $S(F)$ and $S'(F)$, the transfer factor $\Delta_{H,\kappa}$ and the choice of ι) have been checked only for $SL(2)$ and the element $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, where $\Phi^{\kappa}_f(\gamma) = O_\gamma(f) - O_{\gamma'}(f) = f^\vee(\chi_0)$ (§6.2). The description of χ_κ via the n-cycle is correct up to the choice of L-embedding, which may twist it by a central element. Proposition 5.2 assumes a stable version of Theorem 4.5 for H, which is not proved here (for $m < n$, H is not a torus).

6 Worked examples: $PGL(2)$ and $SL(2)$ (new)

In this section $F = \mathbb{Q}_p$ is allowed in the numerical checks ($q = p = 3$), but all formulas hold for any p-adic F. The geometric side is computed on the Bruhat–Tits tree $\mathcal{X}$ (vertices $PGL(2,F)/PGL(2,\mathcal{O})$, regular of degree $q+1$, metric d). The spectral side uses Lemma 2.1 and Lemma 2.2. Every formula below was checked numerically by the scripts `examples/pgl2/pgl2_check.py` and `examples/sl2/sl2_check.py`: orbital integrals by exact counts of vertices, and spectral integrals by the midpoint rule with 4×10^5 nodes on the unit circle.

6.1 $PGL(2)$

Setup. Let $G = PGL(2)$, $K = PGL(2,\mathcal{O})$, A the diagonal torus. Then $X_*(A) = \mathbb{Z}\varpi^\vee$ with $a_n := \varpi^\vee(\pi)^n = \mathrm{diag}(\pi^n, 1)$, and $\alpha^\vee = 2\varpi^\vee$, so $a_\alpha = a_2$. An unramified character is determined by $z = \chi(a_1) \in \mathbb{C}^\times$, unitary iff $|z| = 1$. Then $\chi(a_\alpha) = z^2 =: y$, $W = \{1, s\}$ acts by $z \mapsto z^{-1}$, $A^- = \{a_n : n \geq 0\}$ and $\delta_B^{1/2}(a_n) = q^{-n/2}$. The double cosets are Ka_nK with $\mathrm{vol}(Ka_nK) = (q+1)q^{n-1}$ for $n \geq 1$; let $f_n = \mathbf{1}_{Ka_nK}$. For γ regular elliptic its centralizer S_γ is compact and we give it volume 1, so $O_\gamma(f) = \int_G f(g^{-1}\gamma g)\, dg$.

Spectral side. With $c_\alpha(\chi) = (1 - q^{-1}y)/(1-y)$, Lemma 2.1 reads

$$\Gamma_\chi(a_n) = Q^{-1}q^{-n/2}\big(c_\alpha(\chi^{-1})z^n + c_\alpha(\chi)z^{-n}\big), \qquad Q = 1 + q^{-1},\ n \geq 0, \tag{6.1}$$

and $f_n^\vee(\chi) = \mathrm{vol}(Ka_nK)\Gamma_\chi(a_n)$. On $|z| = 1$, with $z = e^{i\theta}$ and $d\chi = d\theta/2\pi$, Lemma 2.2 gives $d\mu = \rho\, d\chi$ with

$$\rho(z) = \frac{Q}{2}|c_\alpha(\chi)|^{-2} = \frac{Q}{2}\,\frac{|1-y|^2}{|1-q^{-1}y|^2}. \tag{6.2}$$

Plancherel inversion $\int f_n^\vee \Gamma_\chi(a_m)\, d\mu = \delta_{nm}$ holds (checked for $0 \leq n, m \leq 3$); with the 1997 constant $Q = 1+q$ the diagonal entries would be q.

Geometric side. Since f_n is bi-K-invariant, $g^{-1}\gamma g \in Ka_nK$ iff $d(gK, \gamma gK) = n$, so

$$O_\gamma(f_n) = \#\{v \in \mathcal{X} :\ d(v, \gamma v) = n\}. \tag{6.3}$$

Up to conjugacy and the center, a regular elliptic γ generates a quadratic extension $E = F[\gamma]$, and there are three types:

($\mathbf{U}_r$) E/F unramified, $\gamma \in \mathcal{O}_E^\times$ of depth $r \geq 0$. The fixed set of γ is a ball B_r of radius r, and $d(v, \gamma v) = 2m$ at distance $m \geq 1$ from it.

($\mathbf{R}_r$) E/F ramified, $\gamma \in \mathcal{O}_E^\times$ of depth $r \geq 0$. The fixed set is the r-neighborhood of an edge, and $d(v, \gamma v) = 2m$ at distance $m \geq 1$ from it.

($\mathbf{R}'$) E/F ramified and $v_E(\gamma)$ odd. Then γ reverses an edge e, and $d(v, \gamma v) = 2d(v, e) + 1$.

Counting vertices on spheres gives, for $m \geq 1$:

type	$O_\gamma(f_0)$	$O_\gamma(f_{2m})$	$O_\gamma(f_{2m+1})$
(U_r)	$\lvert B_r\rvert = 1 + \frac{(q+1)(q^r-1)}{q-1}$	$(q+1)q^{r+m-1}$	0
(R_r)	$\frac{2(q^{r+1}-1)}{q-1}$	$2q^{r+m}$	0
(R')	0	0	$2q^m$ (and $O_\gamma(f_1) = 2$)

These counts were verified for $q = 3$ by enumerating the 4373 vertices of the ball of radius 7, for $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (U_0), $\left(\begin{smallmatrix} 1 & -3 \\ 3 & 1 \end{smallmatrix}\right)$ (U_1), $\left(\begin{smallmatrix} 1 & 3 \\ 1 & 1 \end{smallmatrix}\right)$ (R_0) and $\left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right)$ (R').

The truncated kernel. If the fixed ball of γ of type (U_r) is centred at the base vertex, then $G(T)$ consists of the vertices at distance $< T$ from it, and (4.4) becomes

$$I^T(\chi, \gamma) = |B_r| + q^r \sum_{m=1}^{M} \big(c_\alpha(\chi^{-1}) y^m + c_\alpha(\chi) y^{-m}\big), \qquad M = T - r - 1, \tag{6.4}$$

because $O_\gamma(f_{2m}) Q^{-1} q^{-m} = q^r$. (Equivalently, $I^T = \sum_{n \leq 2M} O_\gamma(f_n) \Gamma_\chi(a_n)$; this is the truncation used in the separate note `examples/pgl2/pgl2_worked_example.pdf`.) Summing the geometric series gives the vertex decomposition of Proposition 4.2:

$$I^T = I - q^r \Big(c_\alpha(\chi^{-1}) \frac{y^{M+1}}{1-y} + c_\alpha(\chi) \frac{y^{-M-1}}{1-y^{-1}} \Big), \tag{6.5}$$

where $I = I(\chi, \gamma)$ is the T-independent part.

Theorem 6.1. *For $G = PGL(2)$ and regular unitary χ ($z \neq \pm 1$), $I(\chi, \gamma)$ is the rational function*

$$(U_r): \quad I(\chi,\gamma) = |B_r| + q^r \sum_{w \in W} c_\alpha\big((w\chi)^{-1}\big) \frac{(w\chi)(a_\alpha)}{\tilde\theta(w\chi)} = |B_r| + q^r \, \frac{\cos\theta - q^{-1}}{1 - \cos\theta} \qquad (y = e^{i\theta}),$$

$$(R_r): \quad I(\chi,\gamma) = \frac{2(q^{r+1}-1)}{q-1} + \frac{2q^{r+1}}{q+1} \cdot \frac{\cos\theta - q^{-1}}{1-\cos\theta},$$

$$(R'): \quad I(\chi,\gamma) = \frac{q^{1/2}(1-q^{-1})}{q+1} \cdot \frac{\cos\psi}{\sin^2\psi} \qquad (z = e^{i\psi}).$$

Hypothesis (R) holds, and $O_\gamma(f) = \int f^\vee(\chi) I(\chi, \gamma)\, d\mu(\chi)$ for all $f \in \mathcal{H}(G, K)$, the integral converging absolutely.

Proof. Type (U_r); the others are identical. Multiply (6.5) by $\rho = \frac{Q}{2}(c_\alpha(\chi) c_\alpha(\chi^{-1}))^{-1}$:

$$(I^T - I)\rho = -\frac{Q}{2}\, q^r \Big(\frac{y^{M+1}}{1 - q^{-1}y} + \frac{y^{-M-1}}{1 - q^{-1}y^{-1}} \Big).$$

The coefficient of each oscillatory term, times ρ, is smooth on $|z| = 1$, which is (R). Better, the first term has Fourier modes y^k only for $k \geq M+1$ and the second only for $k \leq -M-1$. Since $f^\vee$ is a Laurent polynomial in z of degree $\leq \deg f$ (the largest n with $f(a_n) \neq 0$), $\int f^\vee (I^T - I)\, d\mu = 0$ *exactly* once $2(M+1) > \deg f$. Combine with Theorem 4.5(a). The closed forms follow by summing the geometric series. $I\rho$ is smooth: see the next remark. $\square$

Remark 6.2 (poles and the Plancherel density). $I(\chi,\gamma)$ has a *double* pole at the non-regular characters $z=\pm 1$ (i.e. $y=1$), and no other poles on $|z|=1$. The density ρ has a double zero there, and

$$I(\chi,\gamma)\rho(z)=\frac{Q}{2}\cdot\frac{|B_r|(2-2\cos\theta)+q^r(2\cos\theta-2q^{-1})}{1-2q^{-1}\cos\theta+q^{-2}}\qquad(\text{type U}_r).$$

The cancellation is structural: $c_\alpha((w\chi)^{-1})\rho/\tilde{\theta}(w\chi)=\frac{Q}{2}(1-q^{-1}(w\chi)(a_\alpha))^{-1}$, because $\rho\propto|c|^{-2}$ supplies the factor $1-(w\chi)(a_\alpha)$ that cancels $\tilde{\theta}$. On its own $I(\cdot,\gamma)$ is not locally integrable at $z=\pm1$ against $d\chi$, which may explain why the 1992 abstract called it a "distribution" while the 1997 text called it a "rational function".

Remark 6.3 (the core term, and sign). In type (U_r) the constant $|B_r|=O_\gamma(\mathbf{1}_K)$ comes from the vertices fixed by γ. There no Casselman asymptotic is available, and this is the T-independent remainder J_G of Proposition 3.6, which the 1997 induction could not produce. The second term has the shape predicted by the 1997 Proposition 4.2, a sum over W of c-functions over $\tilde{\theta}$. In types (U_r) and (R_r), I is positive on regular unitary χ (its minimum, at $\cos\theta=-1$, is $|B_r|-q^r\frac{1+q^{-1}}{2}>0$). In type ($\text{R}'$) it changes sign with $\cos\psi$. So $I(\cdot,\gamma)\,d\mu$ is a signed measure in general.

	(U_0)		(U_1)		(R_0)		(R')	
n	tree	spectral	tree	spectral	tree	spectral	tree	spectral
0	1	1.000000	5	5.000000	2	2.000000	0	0.000000
1	0	0.000000	0	0.000000	0	0.000000	2	2.000000
2	4	4.000000	12	12.000000	6	6.000000	0	0.000000
3	0	0.000000	0	0.000000	0	0.000000	6	6.000000
4	12	12.000000	36	36.000000	18	18.000000	0	0.000000
5	0	0.000000	0	0.000000	0	0.000000	18	18.000000
6	36	36.000000	108	108.000000	54	54.000000	0	0.000000
7	0	0.000000	0	0.000000	0	0.000000	54	54.000000

Table 1: $PGL(2)$, $q=3$: $O_\gamma(f_n)$ counted on the tree versus $\int f_n^\vee I(\chi,\gamma)\,d\mu$ with I from Theorem 6.1. For $f=f_4$ and type (U_0), $\int f^\vee I^T\,d\mu$ equals 0 for $2M<4$ and exactly 12 for $2M\geq 4$.

6.2 $SL(2)$: an atom

Setup. Let $G=SL(2)$, $K=SL(2,\mathcal{O})$. Now $X_*(A)=\mathbb{Z}\alpha^\vee$, $a_n=\mathrm{diag}(\pi^n,\pi^{-n})$, $z=\chi(a_1)=\chi(a_\alpha)$, $\delta_B^{1/2}(a_n)=q^{-n}$ and $\mathrm{vol}(Ka_nK)=(q+1)q^{2n-1}$ for $n\geq 1$. Lemma 2.1 reads $\Gamma_\chi(a_n)=Q^{-1}q^{-n}(c(\chi^{-1})z^n+c(\chi)z^{-n})$ with $c(\chi)=(1-q^{-1}z)/(1-z)$, and $\rho=\frac{Q}{2}|1-z|^2/|1-q^{-1}z|^2$. The vertices of G/K are the vertices of $\mathcal{X}$ at even distance from the base vertex v_0. So

$$O_\gamma(f_n)=\#\{v:\ d(v_0,v)\text{ even},\ d(v,\gamma v)=2n\}.$$

The non-regular unitary characters are $z=1$ and $z=-1$. At $z=1$, ρ has a double zero, as for $PGL(2)$. At $z=-1$ (the unramified quadratic character χ_0), $s\chi_0=\chi_0$ but $\chi_0(a_\alpha)=-1\neq 1$, so $\rho(\chi_0)=\frac{Q}{2}\cdot\frac{4}{(1+q^{-1})^2}\neq 0$. Here $i_{G,A}(\chi_0)=\pi^+\oplus\pi^-$ is reducible, with R-group of order 2.

An unramified elliptic element. Let $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (type U_0; it fixes v_0 only), and let γ' be its conjugate by $\mathrm{diag}(\pi, 1) \in GL(2, F)$. Then γ' is stably conjugate but not $SL(2, F)$-conjugate to γ; it fixes only the neighbor $\mathrm{diag}(\pi, 1)v_0$, which has odd type. Counting gives

$$O_\gamma(f_n) = (q+1)q^{n-1} \cdot \tfrac{1+(-1)^n}{2}, \qquad O_{\gamma'}(f_n) = (q+1)q^{n-1} \cdot \tfrac{1-(-1)^n}{2} \qquad (n \geq 1),$$

with $O_\gamma(f_0) = 1$ and $O_{\gamma'}(f_0) = 0$. The paper's truncated kernel (4.4) is

$$I^T(\chi, \gamma) = 1 + \sum_{k=1}^{T-1} \big(c(\chi^{-1})z^{2k} + c(\chi)z^{-2k}\big),$$

since $G(T)$ consists of the vertices at distance $2k < 2T$ from v_0, and $d(v, \gamma v) = 2d(v_0, v)$. Its T-independent part is

$$I(\chi, \gamma) = 1 + c(\chi^{-1})\frac{z^2}{1-z^2} + c(\chi)\frac{z^{-2}}{1-z^{-2}}, \tag{6.6}$$

which is regular on $|z| = 1$ away from $z = 1$. In fact, since $z\,c(\chi^{-1}) + c(\chi) = 1 + z$, a short computation gives the closed form

$$I(\chi, \gamma) = \frac{1 - q^{-1}}{2(1 - \cos\theta)} \qquad (z = e^{i\theta}),$$

which has the same shape as the $PGL(2)$ kernel of type (U_0) in Theorem 6.1. The vertex terms

$$-c(\chi^{-1})\frac{z^{2T}}{1-z^2} - c(\chi)\frac{z^{-2T}}{1-z^{-2}}$$

each have a simple pole at $z = -1$ where $\rho \neq 0$. So Hypothesis (R) *fails*, although their sum is smooth for each T (Remark 2026.9).

Proposition 6.4. *For $G = SL(2)$, γ as above and all $f \in \mathcal{H}(G, K)$, let $I_{st}(\chi) = 1 + c(\chi^{-1})\frac{z}{1-z} + c(\chi)\frac{z^{-1}}{1-z^{-1}}$ (the kernel (6.6) for $SO_\gamma = O_\gamma + O_{\gamma'}$). Then $I(\chi, \gamma) = \frac{1}{2}I_{st}(\chi)$, and*

$$O_\gamma(f) = \int f^\vee(\chi)\, I(\chi, \gamma)\, d\mu(\chi) + \tfrac{1}{2}f^\vee(\chi_0) = \tfrac{1}{2}\int f^\vee(\chi)\, I_{st}(\chi)\, d\mu(\chi) + \tfrac{1}{2}f^\vee(\chi_0),$$

$$O_{\gamma'}(f) = \tfrac{1}{2}\int f^\vee(\chi)\, I_{st}(\chi)\, d\mu(\chi) - \tfrac{1}{2}f^\vee(\chi_0).$$

So $dI_\gamma = I(\chi, \gamma)\, d\mu + \frac{1}{2}\delta_{\chi_0}$: the spectral measure of O_γ has an atom at χ_0, and the κ-orbital integral is

$$\Phi^\kappa_f(\gamma) = O_\gamma(f) - O_{\gamma'}(f) = f^\vee(\chi_0),$$

in agreement with Proposition 5.1 ($n = 2$, $\chi_\kappa = \chi_0$, $\Delta = 1$).

Proof. The stable orbital integral has coefficients $SO_\gamma(f_n) = (q+1)q^{n-1}$ for $n \geq 1$, exactly as for $PGL(2)$ type (U_0), and the same argument as for Theorem 6.1 gives $SO_\gamma(f) = \int f^\vee I_{st}\, d\mu$. The poles of I_{st} are at $z = 1$ only, where ρ has a double zero. For the κ-part, $f_n^\vee(\chi_0) = \mathrm{vol}(Ka_nK)Q^{-1}q^{-n}(-1)^n \cdot 2c(\chi_0) = (q+1)q^{n-1}(-1)^n$ for $n \geq 1$, since $c(\chi_0) = (1+q^{-1})/2 = Q/2$; and $f_0^\vee(\chi_0) = 1$. This equals $O_\gamma(f_n) - O_{\gamma'}(f_n)$. Adding and subtracting gives the formulas for O_γ and $O_{\gamma'}$. Finally, $2I - I_{st} = 1 - \frac{z\,c(\chi^{-1}) + c(\chi)}{1+z} = 0$ by the identity $z\,c(\chi^{-1}) + c(\chi) = 1 + z$, so $I = \frac{1}{2}I_{st}$. □

Remark 2026.14 (10 Nov. 2026) (§6.2). Proposition 6.4 is proved above, and was also verified numerically for $q = 3$ and $f = f_0, \dots, f_3$ (`sl2_check.py`, checks (2), (3) and (5)); the identity $I = \frac{1}{2} I_{st}$ was checked symbolically. For a ramified elliptic $\gamma = \left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ ($E = \mathbb{Q}_3(\sqrt{3})$) there is no atom (check (4)), as the fundamental lemma predicts: the relevant κ is ramified, and κ-orbital integrals of spherical functions vanish.

n	$O_\gamma(f_n)$	$O_{\gamma'}(f_n)$	$\frac{1}{2}\int f_n^\vee I_{st}\, d\mu$	$\frac{1}{2} f_n^\vee(\chi_0)$	$\int f_n^\vee I(\chi,\gamma)\, d\mu$
0	1	0	0.5	0.5	0.5
1	0	4	2.0	−2.0	2.0
2	12	0	6.0	6.0	6.0
3	0	36	18.0	−18.0	18.0

Table 2: $SL(2)$, $q = 3$, $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The atom-free integral in the last column misses exactly $\frac{1}{2} f_n^\vee(\chi_0)$.

7 Route B: the identity from Arthur's local trace formula (new)

This section explains how (1.2) follows from Arthur's local trace formula [Ar91], and serves as an introduction to Arthur's work through this special case. It is a sketch.

Remark 2026.15 (10 Nov. 2026) (§7). Nothing in this section has been checked against [Ar91, Ar93] at the level of normalizations: the constants $|W_0^M||W_0^G|^{-1}$, the signs $(-1)^{\dim(A_M/A_G)}$, the measures on the spectral sets, and the identification of $I(\chi,\phi)$ with a weighted character (Conjecture 7.3). The structural statements (which terms survive) use only the definitions and the splitting formulas.

7.1 Arthur's formula. Let $f = f_1 \otimes f_2 \in C_c^\infty(G(F) \times G(F))$ act on $L^2(G(F))$ by R as in §1, with kernel (1.4). Arthur truncates the diagonal integral with a function $u(x,T)$, the characteristic function of a large compact set modulo A_G built from the building, and shows that $J^T(f) = \int_{G(F)/A_G(F)} K_f(x,x) u(x,T)\, dx$ is asymptotic to a polynomial in T, whose constant term $J(f)$ has two expansions [Ar91]:

$$J(f) = \sum_M \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim(A_M/A_G)} \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma, f)\, d\gamma$$
$$= \sum_M \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim(A_M/A_G)} \int_{\Pi(M)} J_M(\pi, f)\, d\pi.$$

Both sums run over Levi subgroups $M \supseteq M_0$. In the first (geometric) expansion, $\Gamma_{\mathrm{ell}}(M)$ is the set of elliptic conjugacy classes of M, and $J_M(\gamma, f)$ is the weighted orbital integral of $f_1 \otimes f_2$ over γ, with weight $v_M(x_1, x_2)$ given by the (G,M)-family $e^{-\lambda(H_P(x_1) - H_{\bar{P}}(x_2))}$. In the second (spectral) expansion, $\Pi(M)$ is a space of tempered representations of M; in [Ar93] it is reorganized in terms of the virtual elliptic tempered characters $T_{\mathrm{ell}}(M)$. $J_M(\pi, f)$ is the weighted character, built from the (G,M)-family of normalized intertwining operators $\lambda \mapsto R_{P'|P}(\pi)^{-1} R_{P'|P}(\pi_\lambda)$. For $M = G$ the weights are 1 and $J_G(\gamma, f) = |D(\gamma)| O_\gamma(f_1) O_\gamma(f_2)$, $J_G(\pi, f) = \operatorname{tr} \pi(f_1)\, \operatorname{tr} \pi^\vee(f_2)$.

7.2 Specialization: geometric side. Take $f_1 = f \in \mathcal{H}(G,K)$ and $f_2 = \phi$ supported on G_{ell}. By Lemma 4.4, $x \mapsto K_f(x,x)$ has compact support modulo A_G, so $J^T(f) = I(f,\phi)$ for all large T and $J(f) = I(f,\phi)$. For $M \neq G$ and $\gamma \in \Gamma_{\mathrm{ell}}(M)$ regular in G, γ is not elliptic in G, so ϕ vanishes on its G-orbit and $J_M(\gamma, f) = 0$. (Singular γ contribute a set of measure zero.) The geometric side therefore reduces to the $M = G$ term, which is (4.11).

7.3 Specialization: spectral side. The weighted characters satisfy a splitting formula [Ar88]: $J_M(\pi, f_1 \otimes f_2)$ is a sum over pairs (L_1, L_2) of Levi subgroups containing M, with coefficients $d_M^G(L_1, L_2)$, of products of weighted characters of L_1 applied to (a constant term of) f_1 and of L_2 applied to f_2. Two facts simplify it here.

- f_1 *spherical.* Normalized intertwining operators fix the spherical vector of an unramified principal series. So the weighted characters of f_1 for $L_1 \supsetneq M$ vanish, forcing $L_1 = M$ and $L_2 = G$ ($d_M^G(M,G) = 1$). Moreover $\operatorname{tr}\pi(f_1)$ is non-zero only if π has a $K \cap M$-fixed vector, i.e. π is a constituent of an unramified principal series of M.
- ϕ *elliptic.* ϕ is cuspidal: $\operatorname{tr}\sigma(\phi) = 0$ for every properly induced σ. So only the elliptic part of the spectrum survives in the $L_2 = G$ factor, and the factor is the weighted character $J_M(\pi^\vee, \phi)$.

The $M = M_0$ term is therefore

$$\frac{(-1)^{\dim(A_0/A_G)}}{|W|} \int_{\widehat{T}^u} f^\vee(\chi)\, J_{M_0}(\chi^{-1}, \phi)\, d\chi,$$

an absolutely continuous measure in χ. The terms with $M \neq M_0$ come from elliptic tempered virtual characters τ of M with $K \cap M$-fixed vectors. These exist only at unramified χ whose R-group (for M) is non-trivial, and they contribute $f^\vee(\chi)$ times $J_M(\tau^\vee, \phi)$, summed over a set of χ of lower dimension. For $SL(2)$ and $M = G$ this is $\tau = \pi^+ - \pi^-$ at χ_0, with $\operatorname{tr}\tau(f) = f^\vee(\chi_0)$ (the spherical vector lies in π^+) and $\operatorname{tr}\tau^\vee(\phi)$ given by the κ-character. This is the atom of Proposition 6.4.

Conjecture 7.3 (expectation)**.** *For regular unitary χ,*

$$I(\chi,\phi)\rho(\chi) = (-1)^{\dim(A_0/A_G)} |W|^{-1} J_{M_0}(\chi^{-1}, \phi).$$

The singular part dI_ϕ^{sing} of Conjecture 4.6 is the sum of the contributions of the elliptic tempered virtual characters with K-fixed vectors of the Levi subgroups $M \supsetneq M_0$.

7.4 Comparison with Route A. If Arthur's formula is taken as known, §§7.2–7.3 give an unconditional proof of (1.2) in the form of Conjecture 4.6(a): a measure with an absolutely continuous part given by a weighted character and a singular part supported where R-groups are non-trivial. What Route A adds is an explicit and elementary description of the absolutely continuous part, as the T-independent part of the truncated kernel (Proposition 4.2), together with closed forms in rank one (§6). The two truncations are related. Arthur's $u(x,T)$ and our $G(T)$ are both defined by the convex hull of $W \cdot T$ in $\mathfrak{a}_0$. Our vertex terms $\psi(T_Q)R_Q$ (Lemma 3.5A) are the periodic counterparts of Arthur's $e^{\lambda(T_Q)}c_Q(\lambda)/\theta_Q(\lambda)$, and Lemma 2.8 converts one into the other. In rank one, the constant term of $\sum_Q e^{\lambda(T_Q)}c_Q(\lambda)/\theta_Q(\lambda)$ at $\lambda = 0$ is a logarithmic derivative of an intertwining operator, which is how weighted characters arise. That is the structure visible in (6.5).

8 Literature review (new)

The paper was published in 1992, so we divide the literature at that date: §8.1 is the background available when the paper was written, and §8.2 is what came after (including the work that appeared between the published version and the 1997 corrected version).

8.1 Before 1992

Spherical harmonic analysis. The spherical function and the Hecke algebra of a p-adic group go back to Satake [Sa63], who identified $\mathcal{H}(G,K)$ with W-invariant functions on the torus of unramified characters (the Satake isomorphism). Macdonald [Ma71] found the explicit formula for zonal spherical functions (Lemma 2.1) and the spherical Plancherel measure (Lemma 2.2). Casselman [Cas80] re-derived Macdonald's formula from intertwining operators and the asymptotics of matrix coefficients, which are developed in his notes [Cas95] (first circulated in 1974). Cartier's survey [Ca79] collects the unramified theory used throughout this paper. The geometric lemma of Bernstein and Zelevinsky [BZ77] describes Jacquet modules of induced representations. Harish-Chandra's theory of admissible invariant distributions [HC78] supplies the local integrability of characters and the G-admissibility used in §1. Keys [Ke82] determined the R-groups of unramified principal series of Chevalley groups.

Arthur's program. Arthur's trace formula [Ar78] and its invariant form [Ar81, Ar88] introduced truncation, (G,M)-families, and weighted orbital integrals and characters. Arthur [Ar87] expressed characters of supercuspidal representations as weighted orbital integrals, a precursor of the cuspidal specialization in §7.3. "Towards a local trace formula" [Ar89] and "A local trace formula" [Ar91] apply these ideas to the regular representation of $G(F)\times G(F)$ on $L^2(G(F))$, producing the identity used in §7. The elliptic part of harmonic analysis was developed by Kazhdan [Ka86] (cuspidal geometry, the elliptic pairing and density) and Bernstein–Deligne–Kazhdan [BDK86] (trace Paley–Wiener theorem).

Spherical orbital integrals. Explicit spherical orbital integrals appear in Langlands' base change for $GL(2)$ [La80] and in Kottwitz's computation for $GL(3)$ [Ko80]. Waldspurger [Wa89a] computed the spectral expansion of spherical orbital integrals for $GL(N)$, by the method followed here. His Shalika germ paper [Wa89b] and Assem's work on unipotent orbital integrals [As91] give complementary information near the identity.

Endoscopy. Labesse and Langlands [LL79] worked out stable conjugacy, L-indistinguishability and κ-orbital integrals for $SL(2)$; the atom of §6.2 is a shadow of their theory. Kottwitz developed rational and stable conjugacy [Ko82] and the elliptic singular terms of the stable trace formula [Ko86]. Langlands and Shelstad defined transfer factors [LS87]. The fundamental lemma for the unit element of $SL(n)$ was proved by Kazhdan [Ka84] for κ of order n and by Waldspurger [Wa91] in general. Clozel [Cl90] and Labesse [La90] treated stable base change. Henniart's work on the cyclic case of the local Langlands conjecture [He86] uses the κ-orbital integrals of $SL(n)$ considered in §5.

Lattice points. The combinatorial input of §3 is Brion's theorem [Br88] on generating functions of lattice points in rational polytopes.

8.2 After 1992

Elliptic harmonic analysis. Arthur's paper on elliptic tempered characters [Ar93] introduced the virtual characters $T_{\mathrm{ell}}(M)$ and put the local trace formula in its invariant, elliptic form. This is the form used in §7.3, and it explains the singular part of Conjecture 4.6. Waldspurger's local trace formula for p-adic Lie algebras [Wa95a] is the Lie-algebra analogue of [Ar91]. His ICM address [Wa95b] surveys the comparison of orbital integrals as it stood in 1994.

Transfer and the fundamental lemma. Hales [Ha95] reduced the fundamental lemma for the whole Hecke algebra to the unit element. Waldspurger showed that the fundamental lemma implies transfer [Wa97], and Courtès proved transfer for $SL(n)$-type groups [Co97]. Laumon and Ngô proved the fundamental lemma for unitary groups [LN08], and Ngô proved it in general for Lie algebras [Ng10]. Combined with Waldspurger's transfer from Lie algebras to groups and change of characteristic, and with [Ha95], this makes (5.11) a theorem. For statements and surveys, see Hales [Ha05] and Kottwitz [Ko05]. For twisted endoscopy, Lemaire, Mœglin and Waldspurger reduced the fundamental lemma to unit elements [LMW18], and Mœglin and Waldspurger developed the twisted local trace formula [MW18].

Elliptic theory via the building. Schneider and Stuhler [SS97] constructed Euler–Poincaré functions from the Bruhat–Tits building and identified their orbital integrals with characters on the elliptic set. Reeder [Re01] related the Euler–Poincaré pairing of Iwahori-spherical representations to elliptic representations of Weyl groups. Opdam and Solleveld [OS13] computed Ext groups of tempered representations of affine Hecke algebras and recovered Arthur's formula for the elliptic pairing. These results bear directly on the singular part in Conjecture 4.6: the elliptic tempered virtual characters with Iwahori-fixed (in particular K-fixed) vectors are exactly what Reeder and Opdam–Solleveld classify.

The Plancherel formula. Waldspurger's exposition of Harish-Chandra's Plancherel formula [Wa03] gives complete proofs of the harmonic analysis behind §7. A new edition of Macdonald's notes, with A.-M. Aubert, has appeared [MA26].

Local relative trace formulas. The local trace formula has been generalized to relative settings: Feigon [Fe12] and Delorme–Harinck [DH17] for $PGL(2)$, and Beuzart-Plessis [BP20] for the Gan–Gross–Prasad conjecture, following Waldspurger's work on orthogonal groups.

Lattice points. Barvinok's book [Ba08] is a modern account of Brion's theorem and its algorithmic uses.

Remark 2026.16 (10 Nov. 2026) (literature). The bibliographic data below were checked against publisher, Numdam/EuDML or library records in November 2026, except where marked "(not verified)". The review is not exhaustive. In particular, we have not searched systematically for later explicit computations of spherical orbital integrals of $GL(n)$ or $SL(n)$ in terms of the building, which would be the natural comparison for §6.

A Glossary

Unramified group
A reductive group over F that is quasi-split over F and split over an unramified extension. Such a group has hyperspecial maximal compact subgroups.

Hyperspecial maximal compact subgroup K
The stabilizer of a hyperspecial vertex of the building; for G split, a conjugate of $G(\mathcal{O})$ for a smooth reductive $\mathcal{O}$-model. Then $G(F) = P(F)K$ for every parabolic P.

Iwahori subgroup I
The stabilizer of a chamber of the building contained in K; $K = \bigsqcup_w IwI$.

Relative root system; reduced roots
The roots Σ of the maximal split torus A acting on $\operatorname{Lie} G$; reduced means $\alpha/2 \notin \Sigma$.

Coroots, fundamental weights and coweights
$\alpha^\vee \in X_*(A) \otimes \mathbb{Q}$ with $\langle \alpha^\vee, \alpha \rangle = 2$; ϖ_α is the basis dual to $\Delta^\vee$; the fundamental coweights are the basis dual to Δ.

$\mathfrak{a}_P$, $\mathfrak{a}_P^Q$, H_M, $G(F)^1$
$\mathfrak{a}_P = \operatorname{Hom}(X^*(M_P)_F, \mathbb{R})$; $\mathfrak{a}_P^Q$ is the kernel of $\mathfrak{a}_P \to \mathfrak{a}_Q$; $H_M : M(F) \to \mathfrak{a}_M$ is given by $e^{\langle H_M(m), \chi \rangle} = |\chi(m)|$; $G(F)^1 = \ker H_G$.

τ_P^Q, $\hat{\tau}_P^Q$
Characteristic functions of the cones in $\mathfrak{a}_0$ cut out by positivity of the simple roots, resp. of the fundamental weights, of $(M_Q \cap P, A_P)$.

Unramified character; tori $\widehat{T}$, $\widehat{T}^u$
A character of $M_0(F)$ trivial on $M_0(F) \cap K$; equivalently a character of $X_*(A)$. All of them form the complex torus $\widehat{T}$, the unitary ones the compact torus $\widehat{T}^u$, which supports the Plancherel measure.

Regular character
χ with $w\chi \neq \chi$ for $w \neq 1$.

Unramified principal series $i_{G,A}(\chi)$
Normalized induction of χ from B; it has a one-dimensional space of K-fixed vectors, spanned by $\Phi_{K,\chi}$.

Normalized Jacquet functor $r_{M,G}$
$\pi \mapsto \delta_P^{-1/2} \otimes \pi / \langle \pi(n)v - v \rangle$; left adjoint of $i_{G,M}$ (Frobenius reciprocity).

Geometric lemma
The Bernstein–Zelevinsky filtration of $r_{M,G} \circ i_{G,A}(\chi)$ with subquotients $i_{M,A}(w\chi)$, $w \in [W_M \backslash W]$ (Lemma 2.4).

Intertwining operator T_w; c-functions
$T_w : i_{G,A}(\chi) \to i_{G,A}(w\chi)$, given by an integral over $N \cap wN^-w^{-1}$ (convergent in a cone and continued meromorphically). $T_w \Phi_{K,\chi} = c_w(\chi) \Phi_{K,w\chi}$ with c_w a product of rank-one factors c_α (2.8); $c(\chi) = c_{w_\ell}(\chi)$.

Zonal spherical function Γ_χ
The bi-K-invariant matrix coefficient $\langle \pi_\chi(g)\Phi_{K,\chi}, \Phi_{K,\chi^{-1}}\rangle$; normalized by $\Gamma_\chi(1) = 1$.

Macdonald's formula
The explicit formula for Γ_χ on A^- (Lemma 2.1).

Satake (Fourier) transform $f^\vee$
$f^\vee(\chi) = \int f(g)\Gamma_\chi(g^{-1})\,dg$, a W-invariant Laurent polynomial on $\widehat{T}$ (2.9).

Spherical Plancherel measure
$d\mu = \frac{Q_G}{|W|}|c(\chi)|^{-2}\,d\chi$ on $\widehat{T}^u$ (Lemma 2.2); $Q_G = \sum_w (IwI : I)^{-1}$.

Casselman's canonical pairing
The M-invariant pairing between $r_{M,G}(\pi)$ and $r_{M,\bar{P}}(\tilde{\pi})$ that computes matrix coefficients deep in the chamber (3.1).

Level
A vector has level n if it is fixed by the principal congruence subgroup K_n.

Regular elliptic element
A regular semisimple element whose centralizer is an elliptic torus (anisotropic modulo A_G).

Orbital integral
$O_\gamma(f) = \int_{G_\gamma(F)\backslash G(F)} f(g^{-1}\gamma g)\,d\dot{g}$.

Stable conjugacy; stable orbital integral
Elements conjugate over $\bar{F}$ (for $SL(n)$: under $GL(n,F)$); $SO_\gamma(f)$ is the sum of the orbital integrals over the classes in a stable class.

κ-orbital integral
$\sum_x \kappa(x) O_{x^{-1}\gamma x}(f)$ over the classes in a stable class, weighted by a character κ of the group $\mathfrak{D}(S)$ classifying them (5.2).

Weyl discriminant $D(s)$
$\det(1 - \mathrm{Ad}(s))|_{\mathrm{Lie}\, G/\mathrm{Lie}\, S}$.

(G,M)-family; $(G,A)^\sim$-family
Families of functions indexed by $\mathcal{P}(M)$ that agree on common walls; sums $\sum_P c_P/\theta_P$ are then smooth (Lemmas 2.6–2.7). The $\sim$-version is the periodic analogue on $\widehat{T}^u$.

Truncation parameter T; $\mathcal{A}_A(T)$; $G(T)$
$T \in X_*(A)$ regular dominant; $\mathcal{A}_A(T)$ is the set of dominant λ below T in the sense of (3.3), and $G(T) = \bigcup K a_\lambda K$.

Vertex decomposition (Brion)
The generating function of lattice points of a rational polytope is the sum, over its vertices, of the generating functions of the tangent cones (Lemma 3.5A).

Langlands' combinatorial lemma; Γ_M^G
The identity $\sum_{Q \supseteq P} (-1)^{\dim \mathfrak{a}_Q^G} \tau_P^Q \hat{\tau}_Q = \delta_{PG}$ and its relatives [Ar78, CLL84]; used in the 1997 proof of Proposition 3.6.

Hypothesis (R)
Smoothness of the oscillatory coefficients of the truncated kernel times the Plancherel density (§4.3).

R-group
For unramified χ, the quotient of the stabilizer W_χ by its subgroup generated by the reflections s_α with $\chi(a_\alpha) = 1$. It governs the reducibility of $i_{G,A}(\chi)$ for unitary χ.

Elliptic tempered (virtual) character
A virtual tempered character not in the span of properly induced ones (Arthur [Ar93]); for $SL(2)$, $\pi^+ - \pi^-$ at χ_0.

G-admissible distribution
An invariant distribution satisfying Harish-Chandra's finiteness condition [HC78, §14], and hence represented by a locally integrable function on G_{reg}.

Cuspidal function
ϕ with $\operatorname{tr}\sigma(\phi) = 0$ for all properly induced σ; e.g. any ϕ supported on G_{ell}.

Weighted orbital integral; weighted character
Orbital integrals and characters weighted by the (G, M)-families $v_M(x)$, resp. normalized intertwining operators (§7).

Endoscopic group
For $SL(n)$ and κ of order m, $H_m = \ker(N_{E/F} \circ \det) \subset \mathrm{Res}_{E/F} GL(n/m)$, with an L-embedding ${}^L H \to {}^L G$ (§5).

Transfer factor $\Delta_{H,\kappa}$
The Langlands–Shelstad factor relating κ-orbital integrals on G to stable orbital integrals on H [LS87].

Fundamental lemma
The identity (5.11) for spherical functions, $f_H = \iota^* f$; now a theorem.

B Concordance with the 1997 version

Section, equation and result numbers are those of 1997 (new items carry letters or the numbers $6.n$, $7.n$). The item-by-item disposition of the errata log `transcription_log.md` is in `CHANGES.md`. In summary:

1997	2026
Lemma 2.1, 2.2	corrected constant Q_G and chamber (Remark 2026.2)
Lemma 2.4, Cor. 2.5, Remark 1	$[W_M\backslash W]$; "irreducible" deleted; direct-sum statement
§2.5 (Lemmas 2.6–2.9)	kept in place, notation fixed; explanatory role only
(3.1), Lemmas 3.2–3.4	opposite parabolic, $\delta_P^{1/2}$, $A_P^-(\epsilon)$, new proof of 3.2
Lemma 3.5, (3.5)–(3.6), Remark 6	replaced by Lemma 3.5A
(3.7), (3.8)	$Q_a \in \mathbb{Q}$; (3.8) via Lemma 3.5A
Prop. 3.6, proof (3.9)–(3.23)	2026 statement and proof (Remark 2026.8)
Lemma 4.1	kept as a 1997 statement, not used (Remark 2026.9)
Prop. 4.2, 4.3	2026 form; $I(\chi,\phi) = T$-independent part
Theorem 4.5	(a) unconditional, (b) under (R); new Conjecture 4.6
(4.10), (4.11)	Weyl discriminant, measures fixed
§5	rewritten: (5.1)–(5.4A), Props. 5.1–5.2
—	new: §6 examples, §7 Route B, §8 literature, Appendix A

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